

Jerrold E. Marsden and Anthony J. Tromba

Vector Calculus

Fifth Edition

Chapter 7:

Integrals over Paths and Surfaces

7.3 Parametrized Surfaces

7.3 Parametrized Surfaces

Key Points in this Section.

1. To be able to deal with surfaces such as the sphere, one needs to move beyond graphs to more general objects, such as parametrized surfaces.
2. A *parametrized surface* is a map

$$\Phi : D \rightarrow \mathbb{R}^3$$

written as

$$\Phi(u, v) = (x(u, v), y(u, v), z(u, v)).$$

3. The actual surface S is the image of the map Φ .

4. Tangent vectors to the surface are given by

$$\mathbf{T}_u = \frac{\partial \Phi}{\partial u} = \frac{\partial x}{\partial u} \mathbf{i} + \frac{\partial y}{\partial u} \mathbf{j} + \frac{\partial z}{\partial u} \mathbf{k}.$$

and

$$\mathbf{T}_v = \frac{\partial \Phi}{\partial v} = \frac{\partial x}{\partial v} \mathbf{i} + \frac{\partial y}{\partial v} \mathbf{j} + \frac{\partial z}{\partial v} \mathbf{k}.$$

with a normal vector being given by

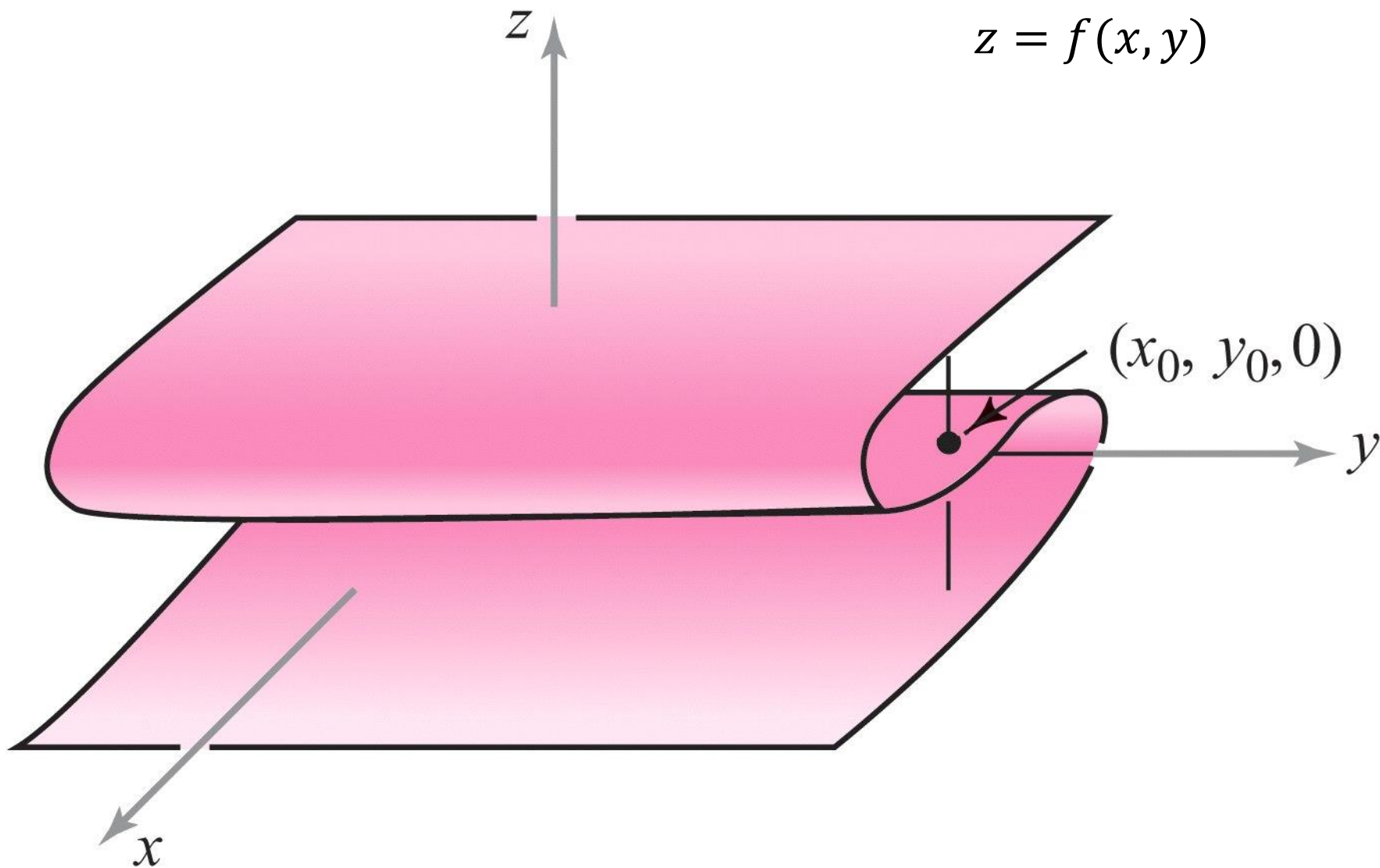
$$\mathbf{n} = \mathbf{T}_u \times \mathbf{T}_v.$$

5. A surface is called *regular* if $\mathbf{T}_u \times \mathbf{T}_v \neq \mathbf{0}$. This nonzero normal vector is useful for finding the equation of the tangent plane to the surface. The tangent plane at a point (x_0, y_0, z_0) on the surface is given by

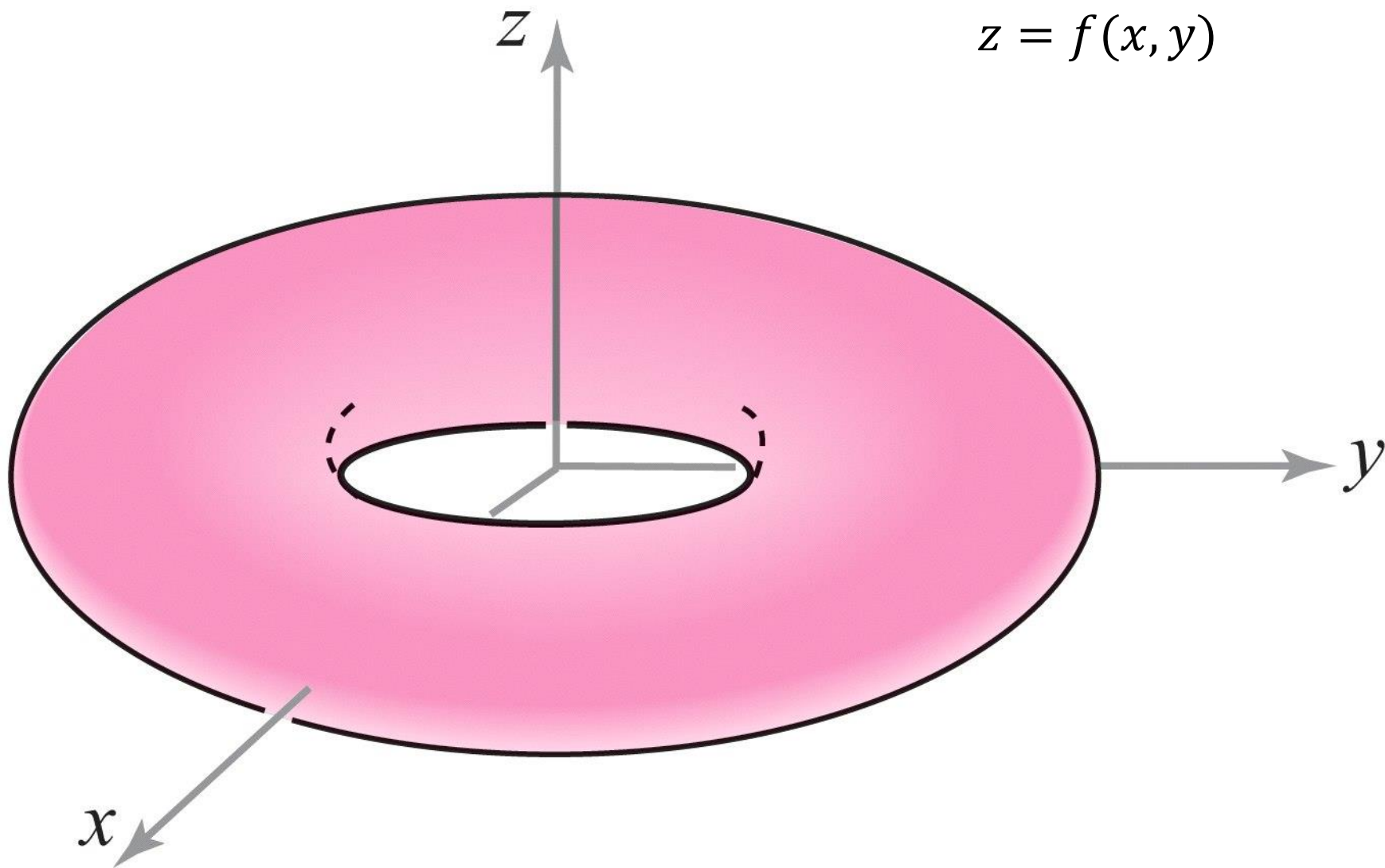
$$(x - x_0, y - y_0, z - z_0) \cdot \mathbf{n} = 0,$$

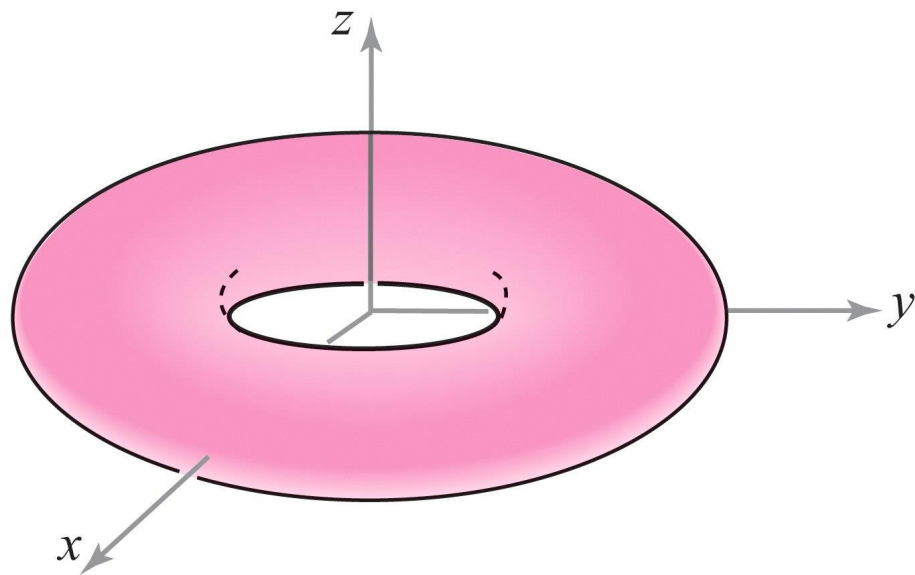
where the normal vector $\mathbf{n}$ is evaluated at the point $(x_0, y_0, z_0) = \Phi(u_0, v_0)$.

Not a function
 $z = f(x, y)$

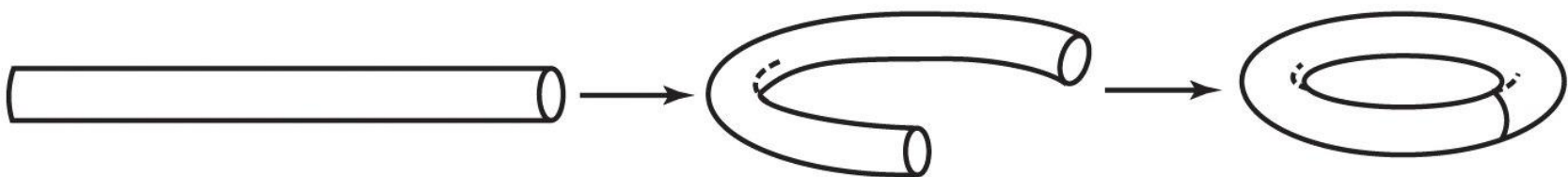
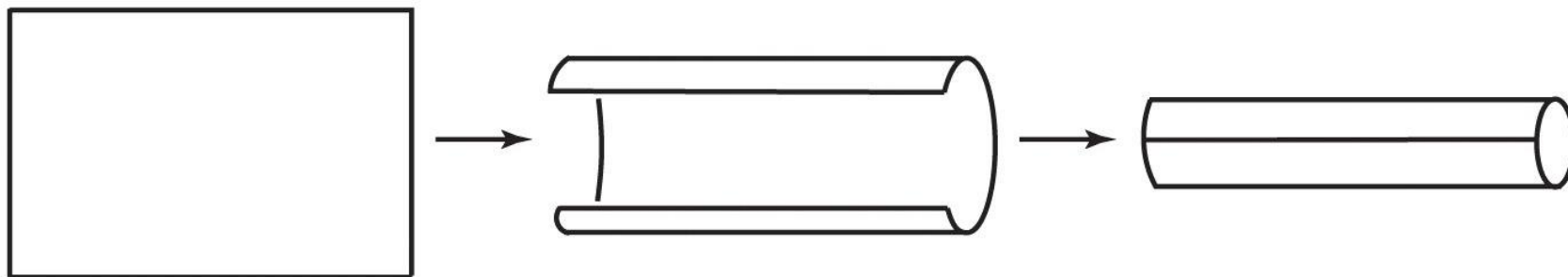


Not a function
 $z = f(x, y)$





Not a function
 $z = f(x, y)$

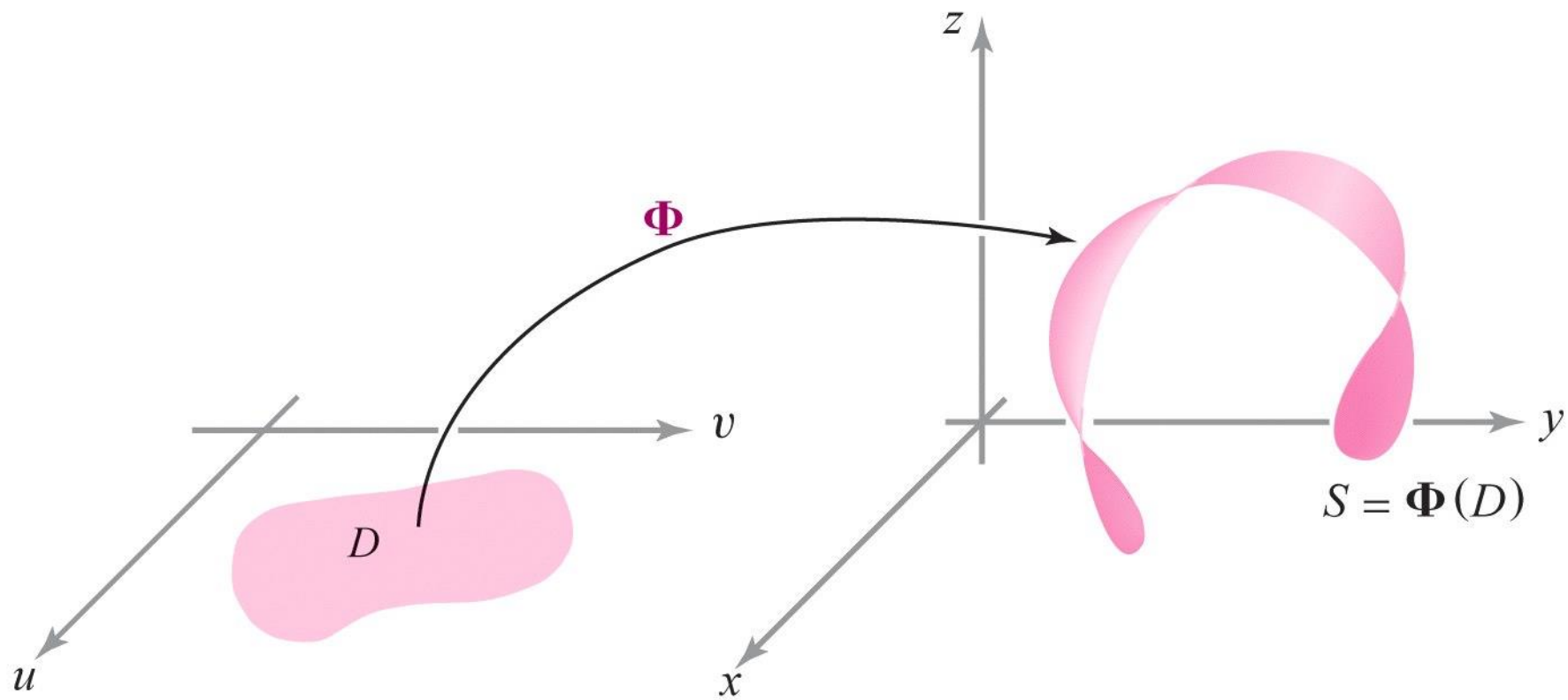


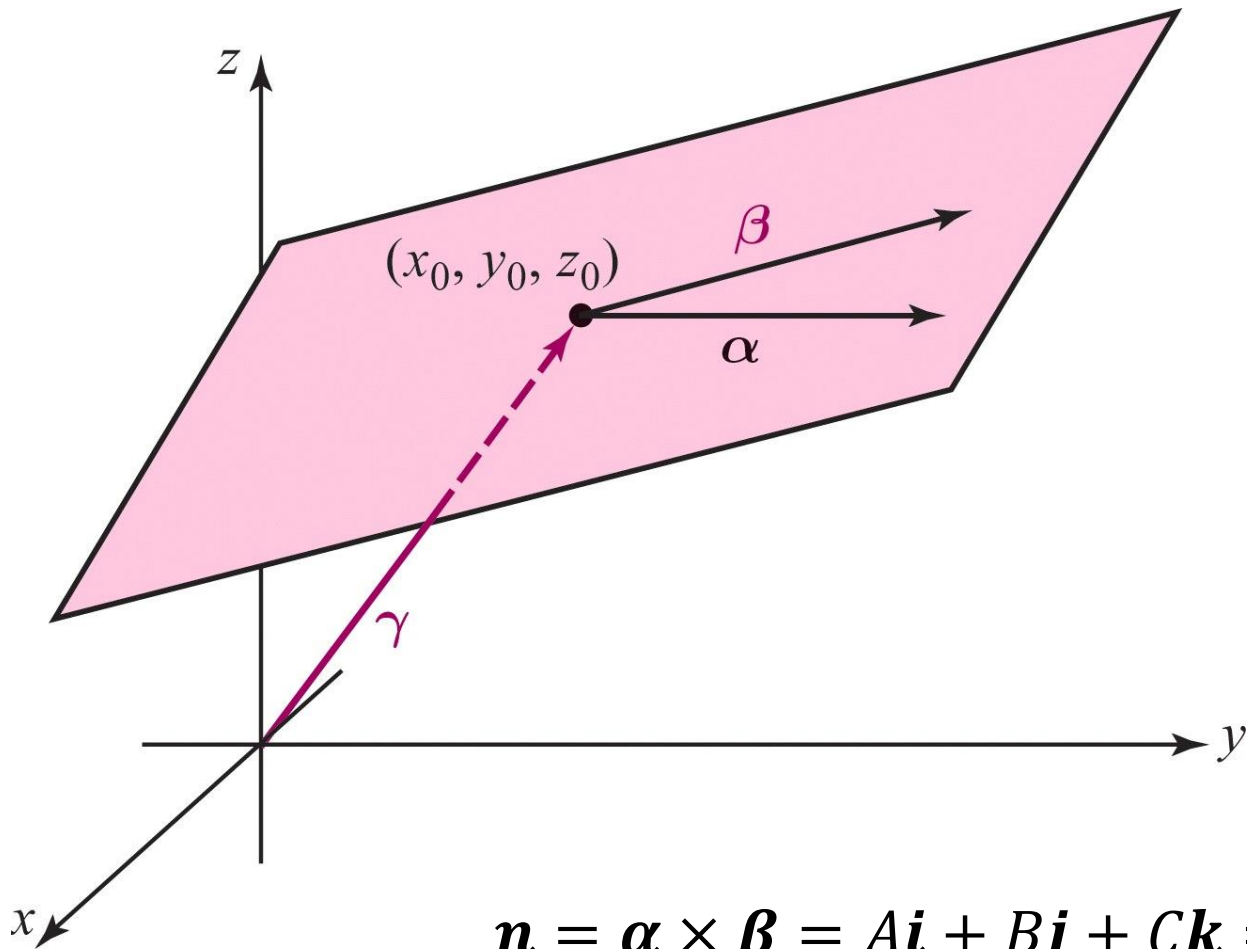
Ends glued

DEFINITION: Parametrized Surfaces A *parametrized surface* is a function $\Phi: D \subset \mathbb{R}^2 \rightarrow \mathbb{R}^3$, where D is some domain in $\mathbb{R}^2$. The *surface* S corresponding to the function Φ is its image: $S = \Phi(D)$. We can write

$$\Phi(u, v) = (x(u, v), y(u, v), z(u, v)).$$

If Φ is differentiable or is of class C^1 [which is the same as saying that $x(u, v)$, $y(u, v)$, and $z(u, v)$ are differentiable or C^1 functions of (u, v)], we call S a *differentiable or a C^1 surface*.

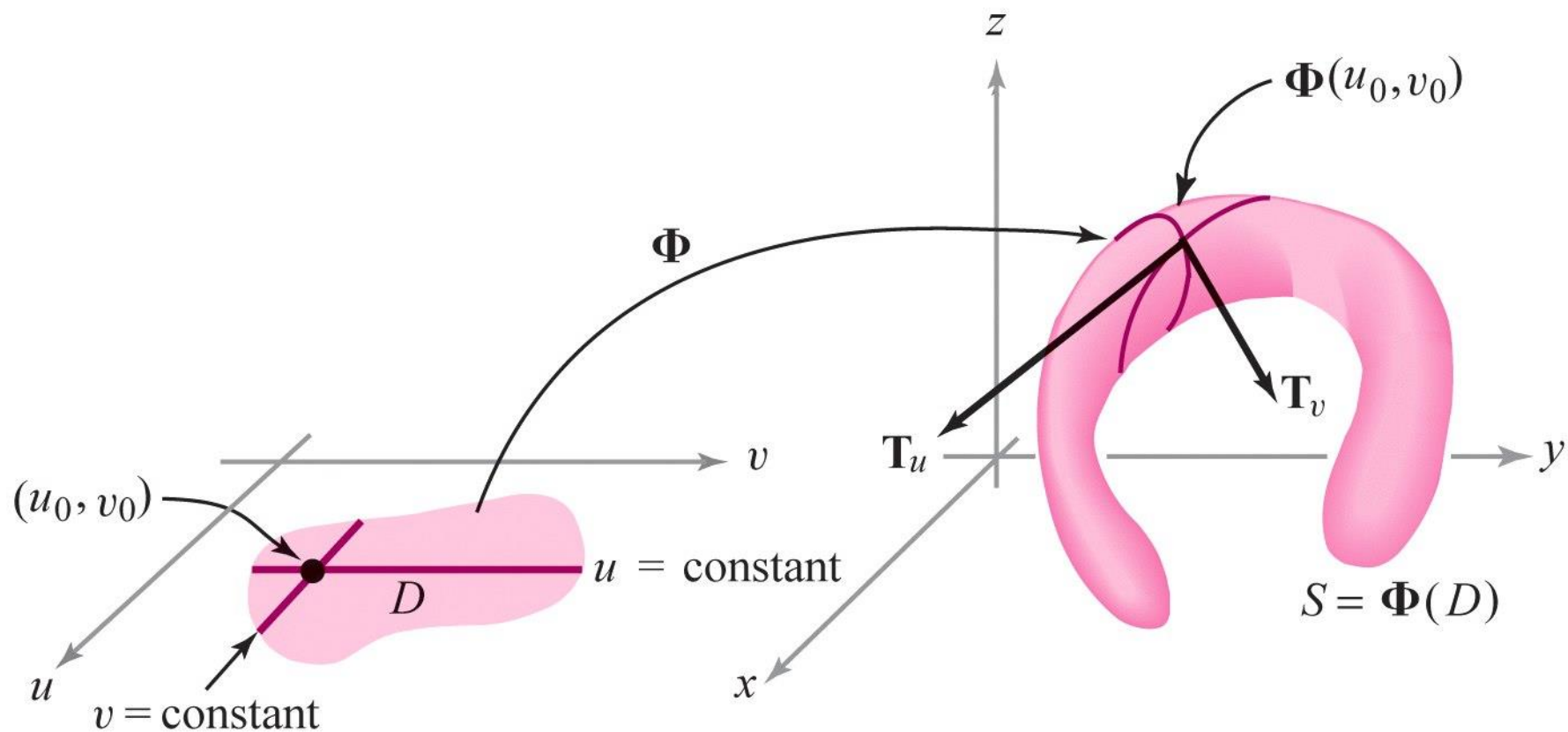




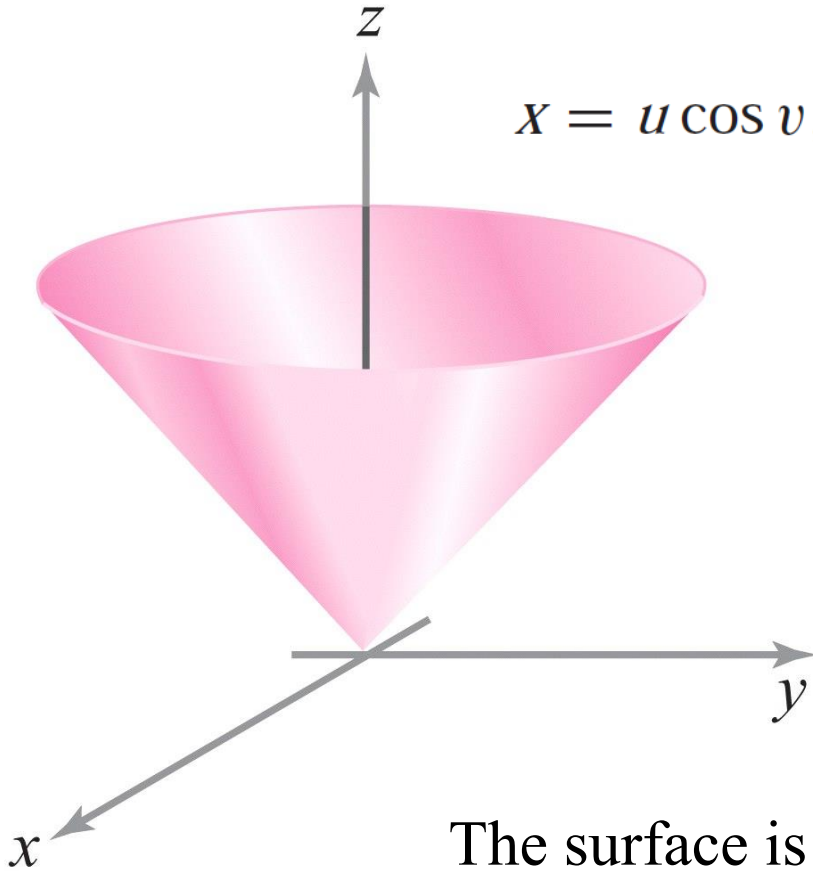
$$\mathbf{n} = \boldsymbol{\alpha} \times \boldsymbol{\beta} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k} = (A, B, C)$$

$$\mathbf{n} \cdot (\mathbf{x} - \mathbf{x}_0) = 0$$

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$



$$x = u \cos v, \quad y = u \sin v, \quad z = u, \quad u \geq 0.$$



The surface is not regular at $(0,0,0)$

$$\mathbf{T}_u = \frac{\partial \Phi}{\partial u} = \frac{\partial x}{\partial u}(0, 0)\mathbf{i} + \frac{\partial y}{\partial u}(0, 0)\mathbf{j} + \frac{\partial z}{\partial u}(0, 0)\mathbf{k} = (\cos 0)\mathbf{i} + (\sin 0)\mathbf{j} + \mathbf{k} = \mathbf{i} + \mathbf{k}$$

$$\mathbf{T}_v = \frac{\partial \Phi}{\partial v} = 0(-\sin 0)\mathbf{i} + 0(\cos 0)\mathbf{j} + 0\mathbf{k} = \mathbf{0}$$

$$\mathbf{T}_u \times \mathbf{T}_v = \mathbf{0}$$

DEFINITION: The Tangent Plane to a Surface If a parametrized surface $\Phi: D \subset \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is *regular* at $\Phi(u_0, v_0)$, that is, if $\mathbf{T}_u \times \mathbf{T}_v \neq \mathbf{0}$ at (u_0, v_0) , we define the *tangent plane* of the surface at $\Phi(u_0, v_0)$ to be the plane determined by the vectors $\mathbf{T}_u$ and $\mathbf{T}_v$. Thus, $\mathbf{n} = \mathbf{T}_u \times \mathbf{T}_v$ is a normal vector, and an equation of the tangent plane at (x_0, y_0, z_0) on the surface is given by

$$(x - x_0, y - y_0, z - z_0) \cdot \mathbf{n} = 0, \quad (1)$$

where $\mathbf{n}$ is evaluated at (u_0, v_0) ; that is, the tangent plane is the set of (x, y, z) satisfying (1). If $\mathbf{n} = (n_1, n_2, n_3) = n_1\mathbf{i} + n_2\mathbf{j} + n_3\mathbf{k}$, then formula (1) becomes

$$n_1(x - x_0) + n_2(y - y_0) + n_3(z - z_0) = 0. \quad (1')$$

Jerrold E. Marsden and Anthony J. Tromba

Vector Calculus

Fifth Edition

Chapter 7:

Integrals over Paths and Surfaces

7.4 Area of a Surface

7.4 Area of a Surface

Key Points in this Section.

1. Area of a parametrized surface:

$$\begin{aligned} A(S) &= \iint_D \|\mathbf{T}_u \times \mathbf{T}_v\| \, du \, dv \\ &= \iint_D \sqrt{\left[\frac{\partial(y, z)}{\partial(u, v)}\right]^2 + \left[\frac{\partial(x, y)}{\partial(u, v)}\right]^2 + \left[\frac{\partial(x, z)}{\partial(u, v)}\right]^2} \, du \, dv \end{aligned}$$

2. The *scalar surface area element* is the integrand:

$$\begin{aligned} dS &= \|\mathbf{T}_u \times \mathbf{T}_v\| \, du \, dv \\ &= \sqrt{\left[\frac{\partial(y, z)}{\partial(u, v)}\right]^2 + \left[\frac{\partial(x, y)}{\partial(u, v)}\right]^2 + \left[\frac{\partial(x, z)}{\partial(u, v)}\right]^2} \, du \, dv \end{aligned}$$

3. The formula for the area element is motivated by the fact that on a small patch, the surface is approximated by the parallelogram with sides $\mathbf{T}_u du$ and $\mathbf{T}_v dv$ and the fact that the area of a parallelogram with sides $\mathbf{a}$ and $\mathbf{b}$ is given by $\|\mathbf{a} \times \mathbf{b}\|$.

4. **Sphere.** $x^2 + y^2 + z^2 = R^2$, the scalar surface element is given by:

$$dS = R^2 \sin \phi \, d\phi \, d\theta$$

5. **Graph.** $z = g(x, y)$ (where $(x, y) \in D \subset \mathbb{R}^2$ can be parametrized by

$$x = u, \quad y = v, \quad z = g(u, v).$$

6. **Surface area of a graph.**

$$A(S) = \iint_D \left(\sqrt{\left[\frac{\partial g}{\partial x} \right]^2 + \left[\frac{\partial g}{\partial y} \right]^2 + 1} \right) du \, dv$$

7. Surfaces of Revolution.

(a) Revolve $y = f(x)$, where $a \leq x \leq b$, about the x -axis:

$$A(S) = 2\pi \int_a^b \left(|f(x)| \sqrt{1 + (f'(x))^2} \right) dx$$

(b) Revolve $y = f(x)$, where $a \leq x \leq b$, about the y -axis:

$$A(S) = 2\pi \int_a^b \left(|x| \sqrt{1 + (f'(x))^2} \right) dx$$

8. The formulas in points 6 and 7 are derived from the general area formula in point 1 for a parametrized surface by parametrizing the circles making up the surface using sines and cosines.

DEFINITION: Area of a Parametrized Surface We define the *surface area*¹⁰ $A(S)$ of a parametrized surface by

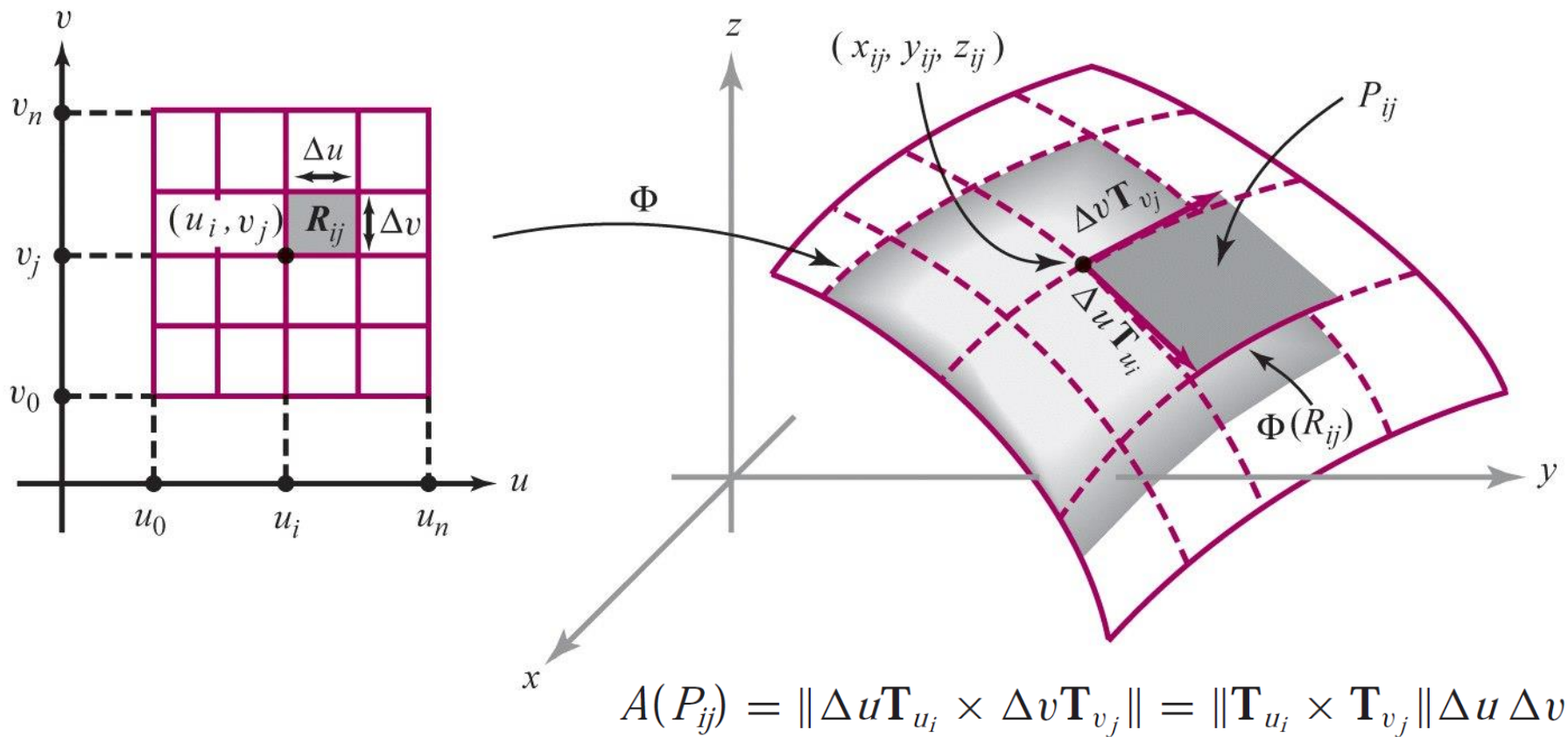
$$A(S) = \iint_D \|\mathbf{T}_u \times \mathbf{T}_v\| \, du \, dv \quad (1)$$

where $\|\mathbf{T}_u \times \mathbf{T}_v\|$ is the norm of $\mathbf{T}_u \times \mathbf{T}_v$. If S is a union of surfaces S_i , its area is the sum of the areas of the S_i .

$$\mathbf{T}_u = \frac{\partial x}{\partial u}(u, v)\mathbf{i} + \frac{\partial y}{\partial u}(u, v)\mathbf{j} + \frac{\partial z}{\partial u}(u, v)\mathbf{k}$$

$$\mathbf{T}_v = \frac{\partial x}{\partial v}(u, v)\mathbf{i} + \frac{\partial y}{\partial v}(u, v)\mathbf{j} + \frac{\partial z}{\partial v}(u, v)\mathbf{k}$$

$$\|\mathbf{T}_u \times \mathbf{T}_v\| = \sqrt{\left[\frac{\partial(x, y)}{\partial(u, v)}\right]^2 + \left[\frac{\partial(y, z)}{\partial(u, v)}\right]^2 + \left[\frac{\partial(x, z)}{\partial(u, v)}\right]^2}$$



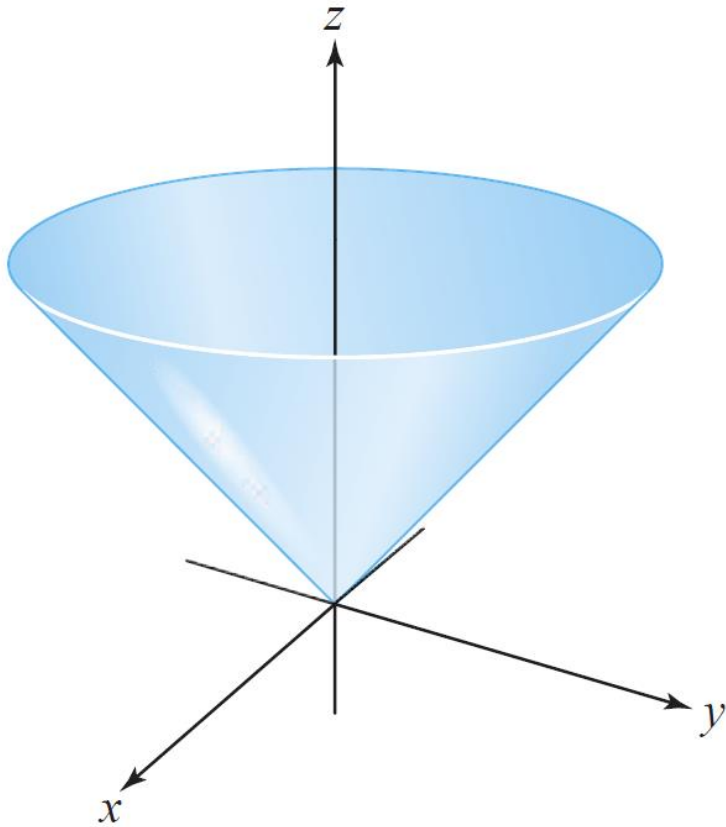
$$A_n = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} A(P_{ij}) = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \|\mathbf{T}_{u_i} \times \mathbf{T}_{v_j}\| \Delta u \Delta v$$

$$A(S) = \lim_{n \rightarrow \infty} A_n = \iint_D \|\mathbf{T}_u \times \mathbf{T}_v\| \, du \, dv$$

Let D be the region determined by $0 \leq \theta \leq 2\pi, 0 \leq r \leq 1$ and let the function $\Phi: D \rightarrow \mathbb{R}^3$, defined by

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = r,$$

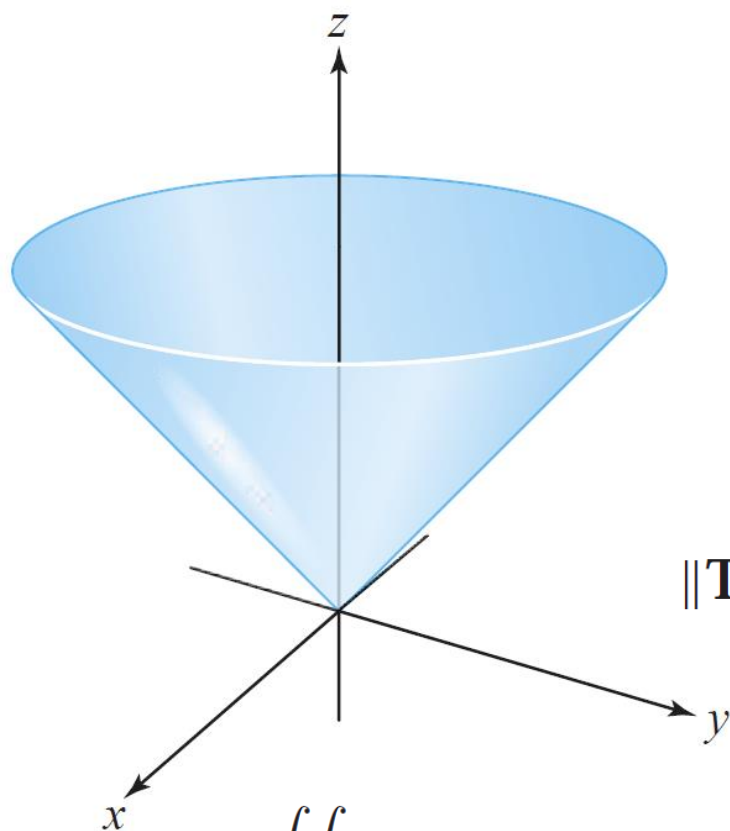
be a parametrization of a cone S (see Figure 7.3.8). Find its surface area.



Let D be the region determined by $0 \leq \theta \leq 2\pi, 0 \leq r \leq 1$ and let the function $\Phi: D \rightarrow \mathbb{R}^3$, defined by

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = r,$$

be a parametrization of a cone S (see Figure 7.3.8). Find its surface area.



$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r,$$

$$\frac{\partial(y, z)}{\partial(r, \theta)} = \begin{vmatrix} \sin \theta & r \cos \theta \\ 1 & 0 \end{vmatrix} = -r \cos \theta,$$

$$\frac{\partial(x, z)}{\partial(r, \theta)} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ 1 & 0 \end{vmatrix} = r \sin \theta,$$

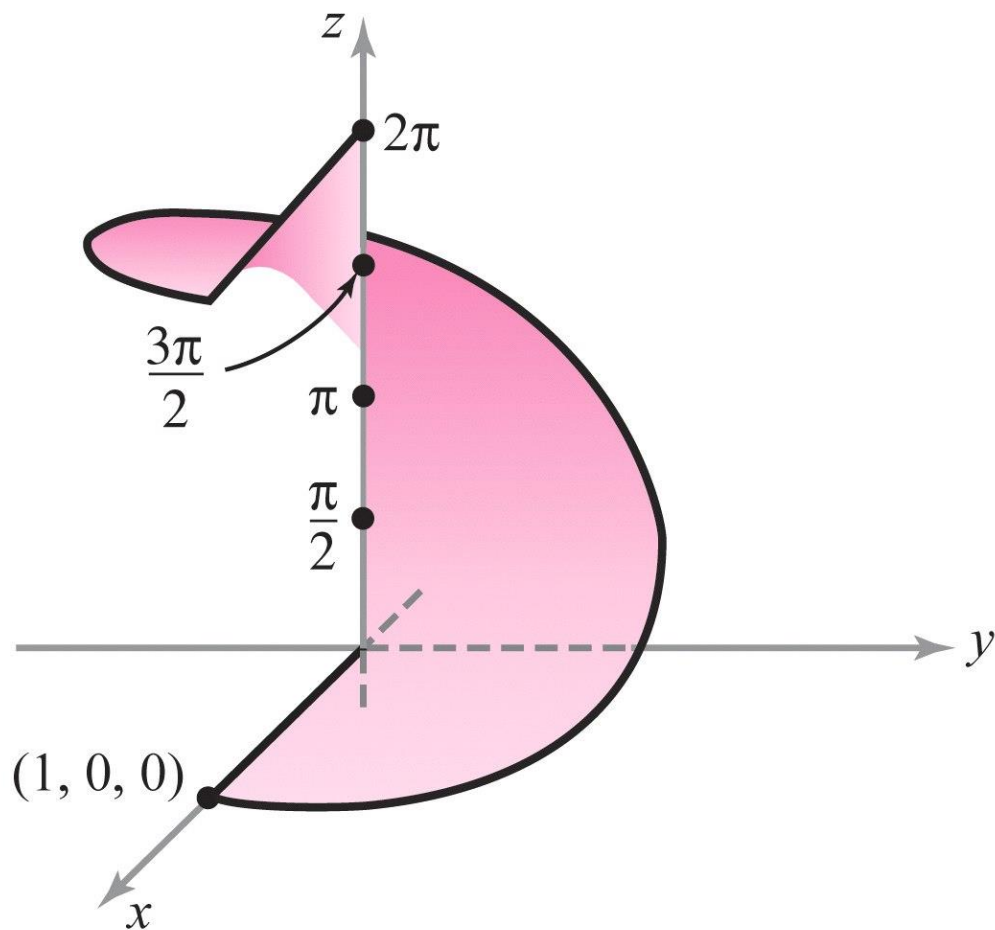
$$\|\mathbf{T}_r \times \mathbf{T}_\theta\| = \sqrt{r^2 + r^2 \cos^2 \theta + r^2 \sin^2 \theta} = r\sqrt{2}.$$

$$\iint_D \|\mathbf{T}_r \times \mathbf{T}_\theta\| \, dr \, d\theta = \int_0^{2\pi} \int_0^1 \sqrt{2}r \, dr \, d\theta = \int_0^{2\pi} \frac{1}{2}\sqrt{2} \, d\theta = \sqrt{2}\pi$$

A *helicoid* is defined by $\Phi: D \rightarrow \mathbb{R}^3$, where

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = \theta$$

and D is the region where $0 \leq \theta \leq 2\pi$ and $0 \leq r \leq 1$ (Figure 7.4.2). Find its area.



A **helicoid** is defined by $\Phi: D \rightarrow \mathbb{R}^3$, where

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = \theta$$

and D is the region where $0 \leq \theta \leq 2\pi$ and $0 \leq r \leq 1$ (Figure 7.4.2). Find its area.

We compute $\partial(x, y)/\partial(r, \theta) = r$ as in Example 1, and

$$\begin{aligned} \frac{\partial(y, z)}{\partial(r, \theta)} &= \begin{vmatrix} \sin \theta & r \cos \theta \\ 0 & 1 \end{vmatrix} = \sin \theta, \\ \frac{\partial(x, z)}{\partial(r, \theta)} &= \begin{vmatrix} \cos \theta & -r \sin \theta \\ 0 & 1 \end{vmatrix} = \cos \theta. \end{aligned}$$

The area integrand is therefore $\sqrt{r^2 + 1}$, which never vanishes, so the surface is regular. The area of the helicoid is

$$\iint_D \|\mathbf{T}_r \times \mathbf{T}_\theta\| \, dr \, d\theta = \int_0^{2\pi} \int_0^1 \sqrt{r^2 + 1} \, dr \, d\theta = 2\pi \int_0^1 \sqrt{r^2 + 1} \, dr.$$

After a little computation (using the table of integrals), we find that this integral is equal to

$$\pi[\sqrt{2} + \log(1 + \sqrt{2})]. \quad \blacktriangle$$

Surface Area of a Graph

A surface S given in the form $z = g(x, y)$, where $(x, y) \in D$, admits the parametrization

$$x = u, \quad y = v, \quad z = g(u, v)$$

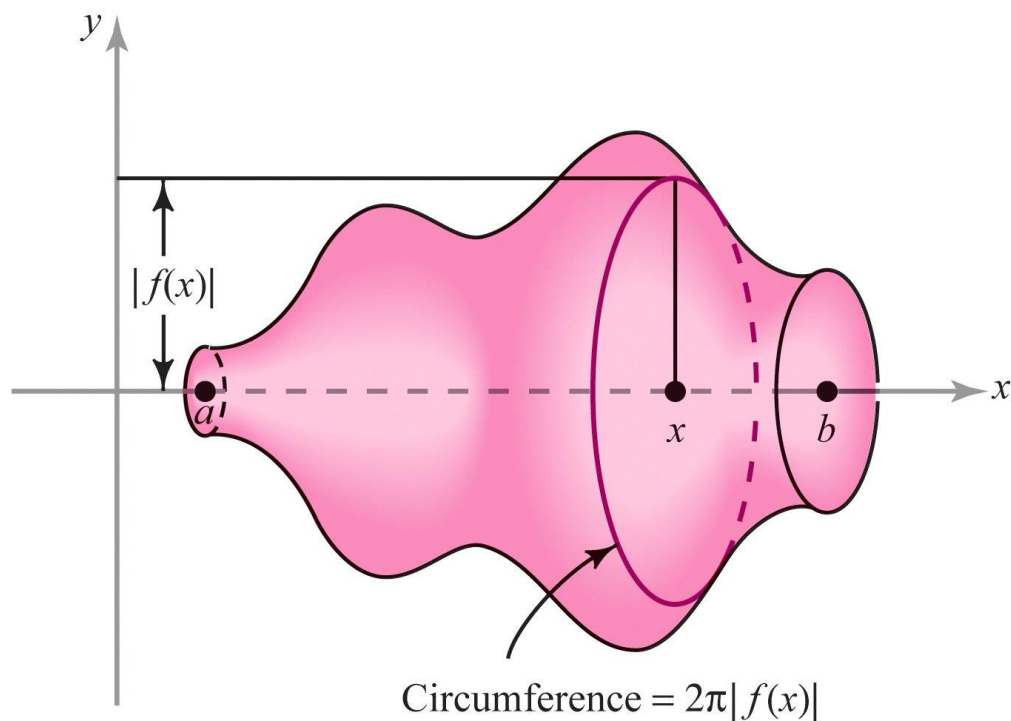
for $(u, v) \in D$. When g is of class C^1 , this parametrization is smooth, and the formula for surface area reduces to

$$A(S) = \iint_D \left(\sqrt{\left(\frac{\partial g}{\partial x} \right)^2 + \left(\frac{\partial g}{\partial y} \right)^2 + 1} \right) dA, \quad (4)$$

Surfaces of Revolution

Surface obtained by revolving function $y = f(x)$ around x axis

$$A = 2\pi \int_a^b (|f(x)| \sqrt{1 + [f'(x)]^2}) dx$$



$$x = u$$

$$y = f(u) \cos v$$

$$z = f(u) \sin v$$

$$a \leq u \leq b$$

$$0 \leq v \leq 2\pi$$

Surfaces of Revolution

$$\frac{\partial(x, y)}{\partial(u, v)} = -f(u) \sin v, \quad \frac{\partial(y, z)}{\partial(u, v)} = f(u) f'(u), \quad \frac{\partial(x, z)}{\partial(u, v)} = f(u) \cos v,$$

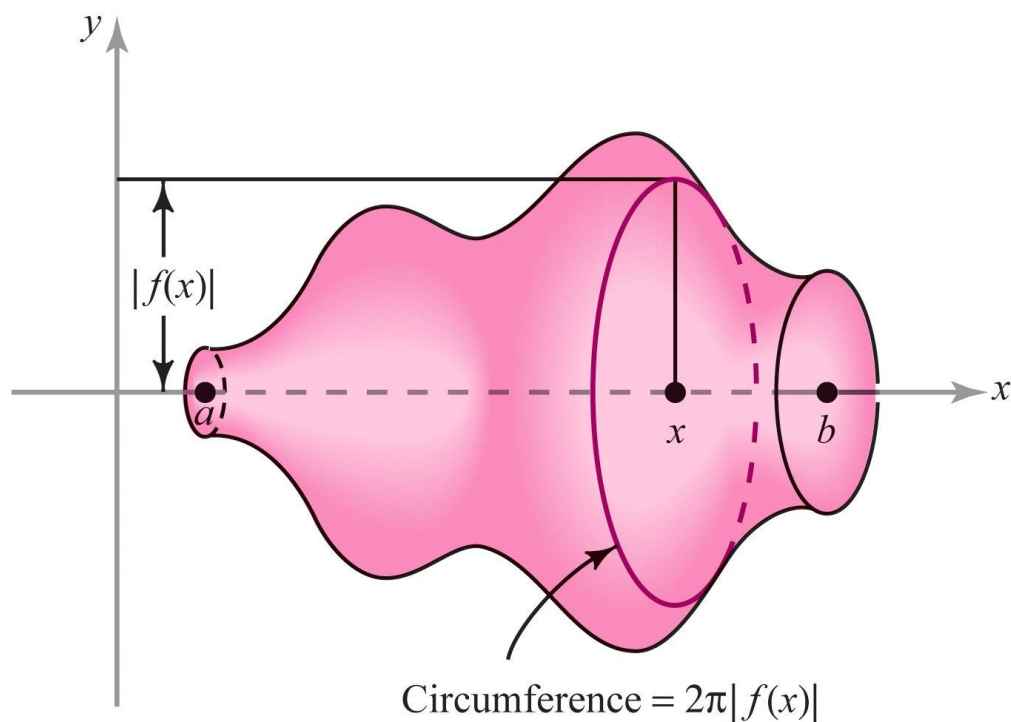
$$\begin{aligned} A(S) &= \iint_D \sqrt{\left[\frac{\partial(x, y)}{\partial(u, v)} \right]^2 + \left[\frac{\partial(y, z)}{\partial(u, v)} \right]^2 + \left[\frac{\partial(x, z)}{\partial(u, v)} \right]^2} du dv \\ &= \iint_D \sqrt{[f(u)]^2 \sin^2 v + [f(u)]^2 [f'(u)]^2 + [f(u)]^2 \cos^2 v} du dv \\ &= \iint_D |f(u)| \sqrt{1 + [f'(u)]^2} du dv \\ &= \int_a^b \int_0^{2\pi} |f(u)| \sqrt{1 + [f'(u)]^2} dv du \\ &= 2\pi \int_a^b |f(u)| \sqrt{1 + [f'(u)]^2} du, \end{aligned}$$

Surfaces of Revolution

Surface obtained by revolving function $y = f(x)$ around x axis

$$A = 2\pi \int_a^b (|f(x)| \sqrt{1 + [f'(x)]^2}) dx$$

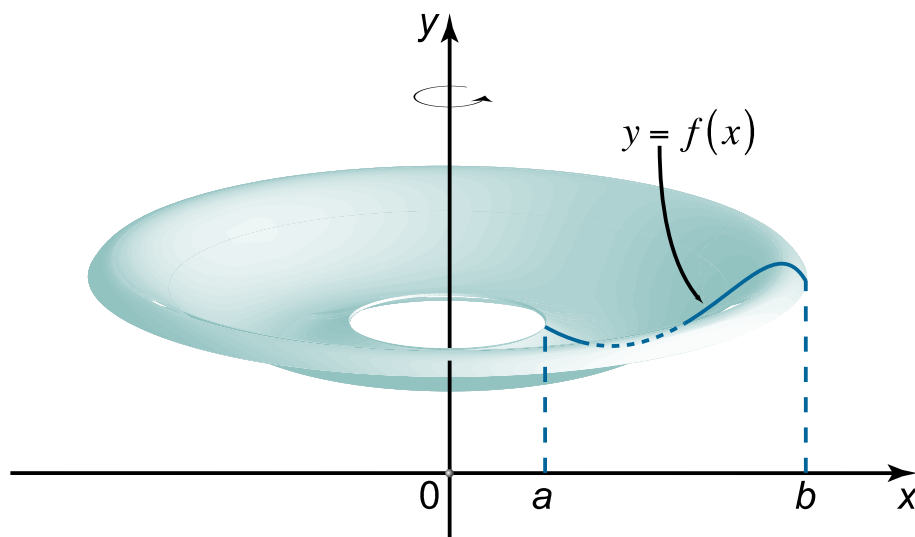
$$= \int_c 2\pi |f(x)| ds$$



Surfaces of Revolution

Surface obtained by revolving function $y = f(x)$ around y axis

$$A = 2\pi \int_a^b (|x| \sqrt{1 + [f'(x)]^2}) dx$$



$$x = u \cos v$$

$$y = u \sin v$$

$$z = f(u)$$

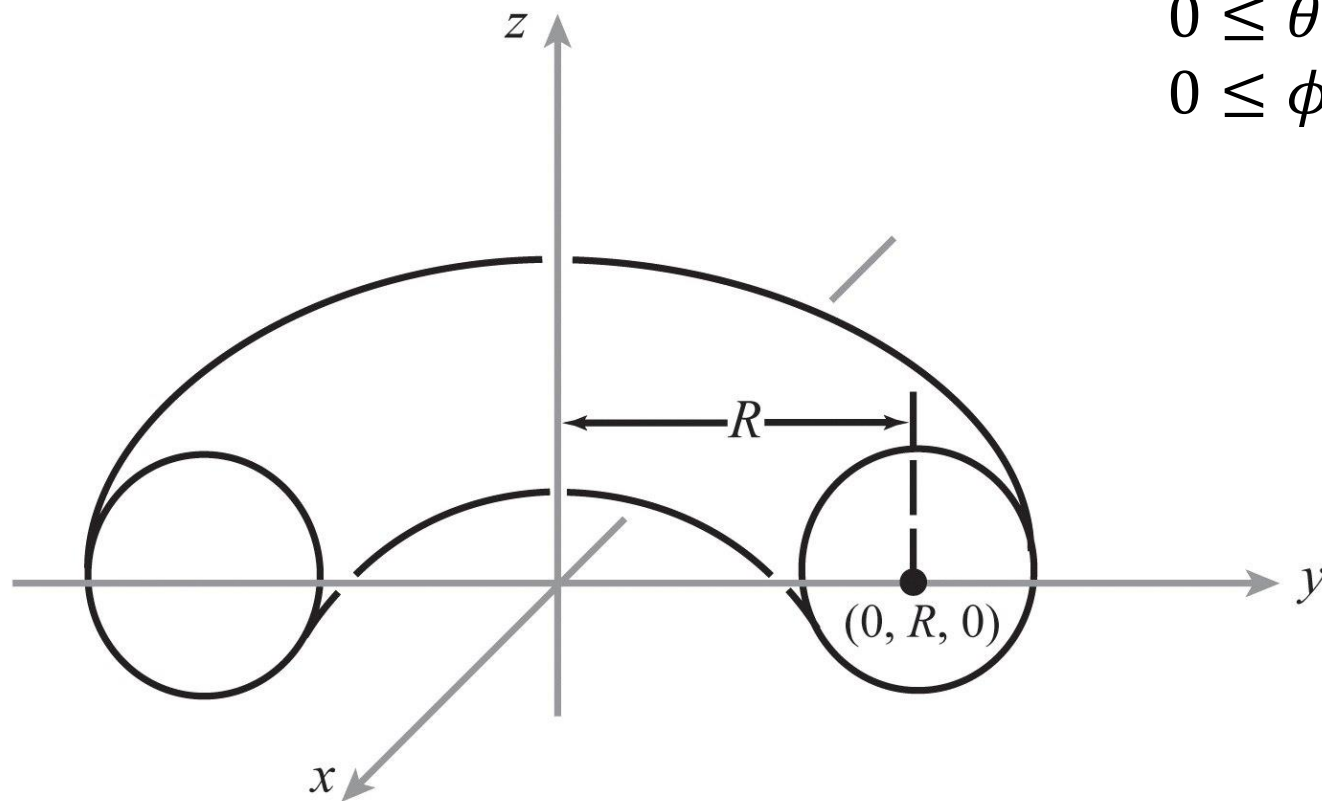
$$a \leq u \leq b$$

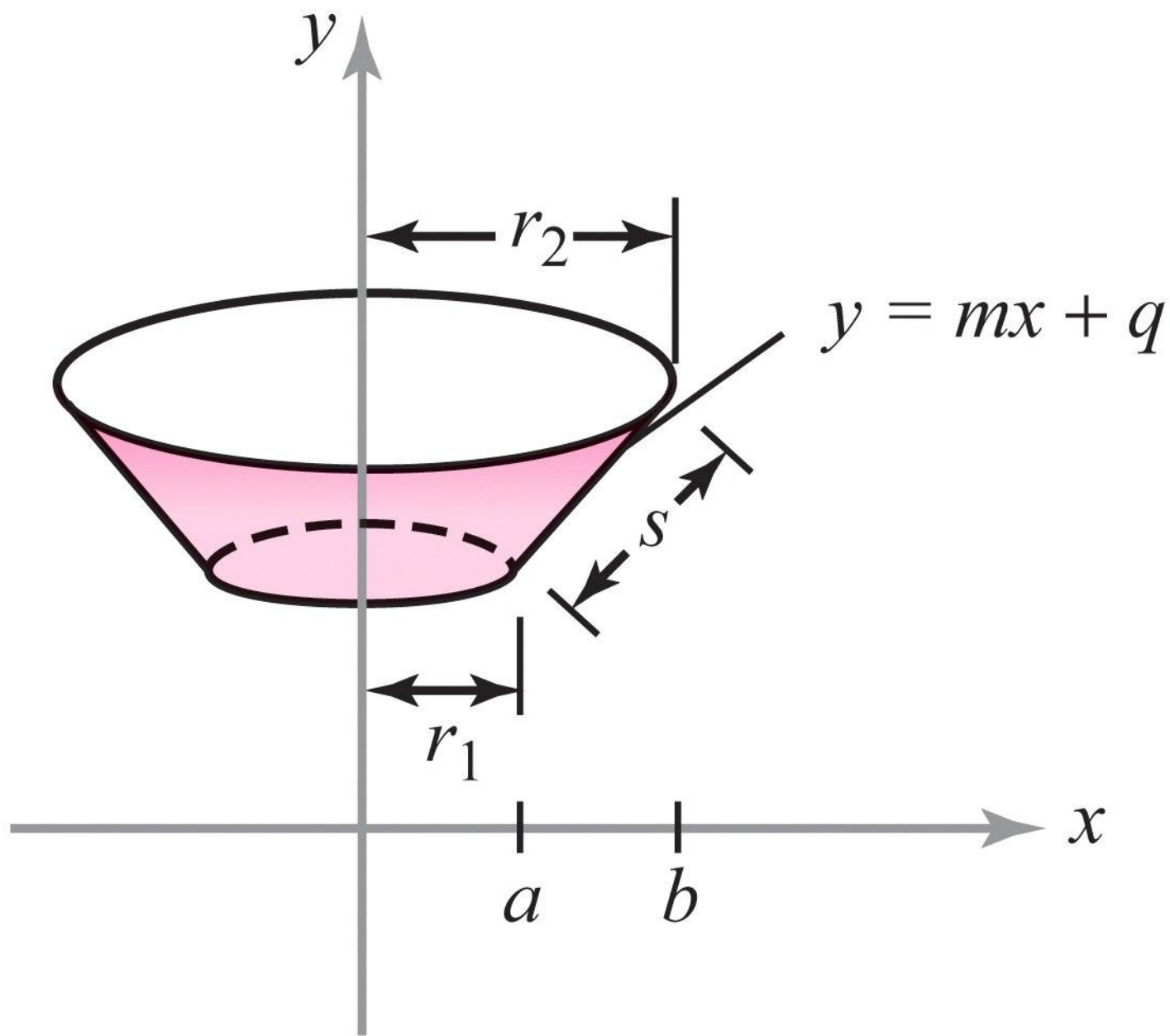
$$0 \leq v \leq 2\pi$$

$$\begin{aligned}x &= (R + \cos \phi) \cos \theta \\y &= (R + \cos \phi) \sin \theta \\z &= \sin \phi\end{aligned}$$

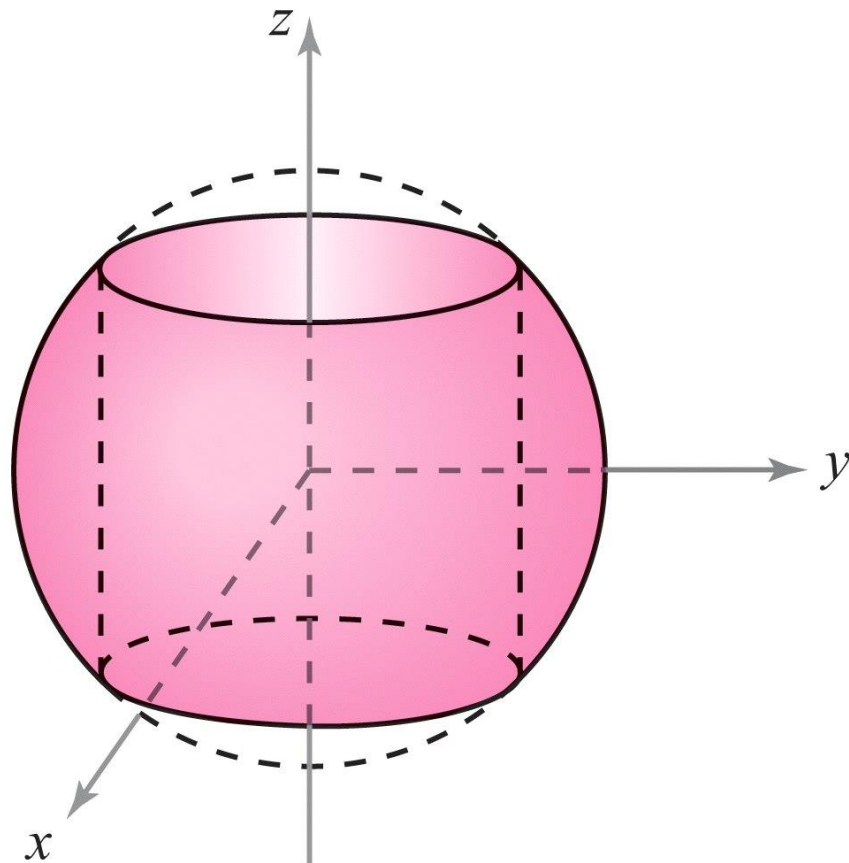
$$0 \leq \theta \leq 2\pi$$

$$0 \leq \phi \leq 2\pi$$





A cylindrical hole of radius 1 is bored through a solid ball of radius 2 to form a ring coupler, as shown in Figure 7.4.8. Find the volume and outer surface area of this coupler.



Jerrold E. Marsden and Anthony J. Tromba

Vector Calculus

Fifth Edition

Chapter 7:

Integrals over Paths and Surfaces

7.5 Integrals of Scalar Functions Over Surfaces

7.5 Integrals of Scalar Functions over Surfaces

Key Points in this Section.

1. **Definition of Scalar Surface Integral.**

$$\iint_S f \, dS = \iint_D f(x(u, v), y(u, v), z(u, v)) \|\mathbf{T}_u \times \mathbf{T}_v\| \, du \, dv$$

2. **Graph.** For $z = g(x, y)$ with $\Phi(u, v) = (u, v, g(u, v))$,

$$\mathbf{T}_u = \mathbf{i} + \frac{\partial g}{\partial u} \mathbf{k}; \quad \mathbf{T}_v = \mathbf{j} + \frac{\partial g}{\partial v} \mathbf{k}$$

and

$$\mathbf{T}_u \times \mathbf{T}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & \frac{\partial g}{\partial u} \\ 0 & 1 & \frac{\partial g}{\partial v} \end{vmatrix} = -\frac{\partial g}{\partial u} \mathbf{i} - \frac{\partial g}{\partial v} \mathbf{j} + \mathbf{k}$$

3. Scalar Surface Element Formulas.

(a) Parametrized Surface.

$$dS = \|\mathbf{T}_u \times \mathbf{T}_v\| \, du \, dv$$

(b) Graph.

$$dS = \frac{dx dy}{\cos \theta} = \frac{dx \, dy}{\mathbf{n} \cdot \mathbf{k}} = \left(\sqrt{\left(\frac{\partial g}{\partial x} \right)^2 + \left(\frac{\partial g}{\partial y} \right)^2 + 1} \right) dx \, dy$$

where $\cos \theta = \mathbf{n} \cdot \mathbf{k}$, and $\mathbf{n}$ is the upward pointing unit normal vector to the surface. See Figure 7.5.1.

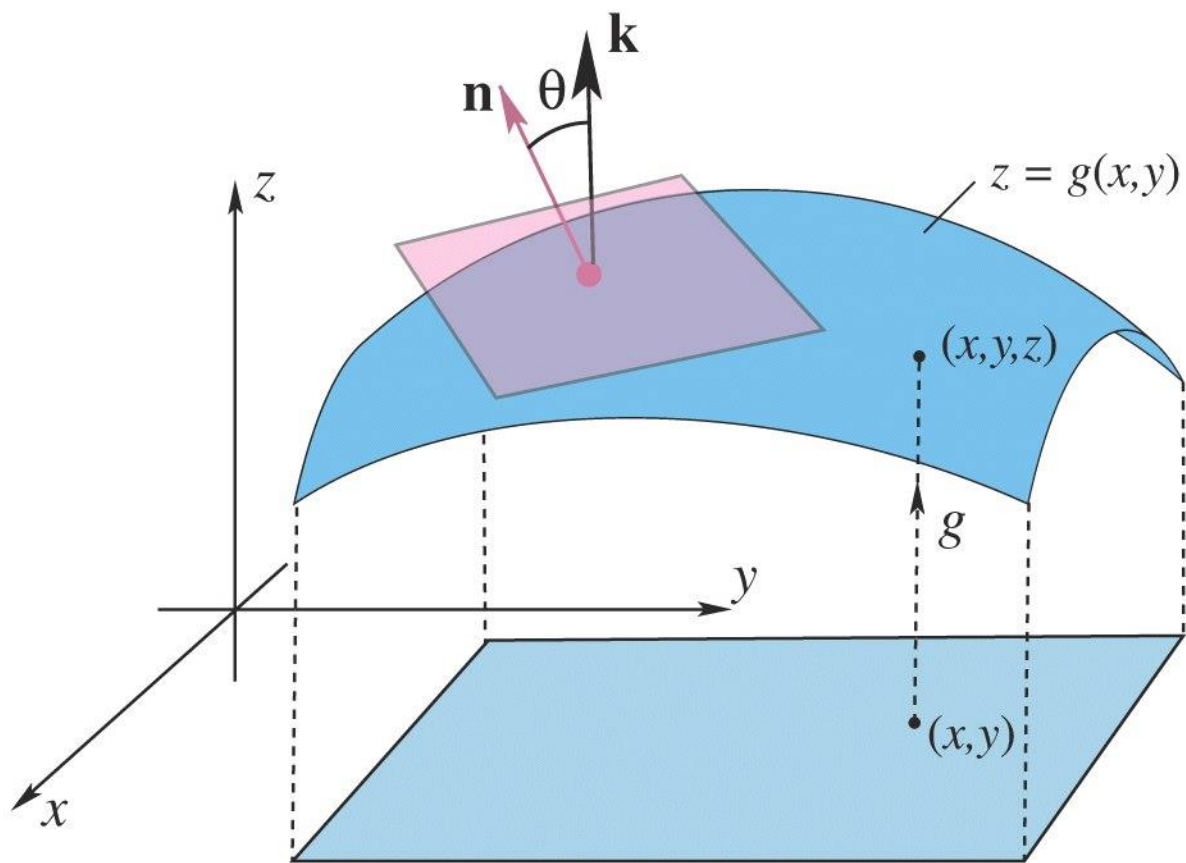


FIGURE 7.5.1. The area element on a graph is $dS = \frac{dx dy}{\cos \theta} = \frac{dx dy}{\mathbf{n} \cdot \mathbf{k}}$.

(c) **Sphere** $x^2 + y^2 + z^2 = R^2$:

$$dS = R^2 \sin \phi d\phi d\theta$$

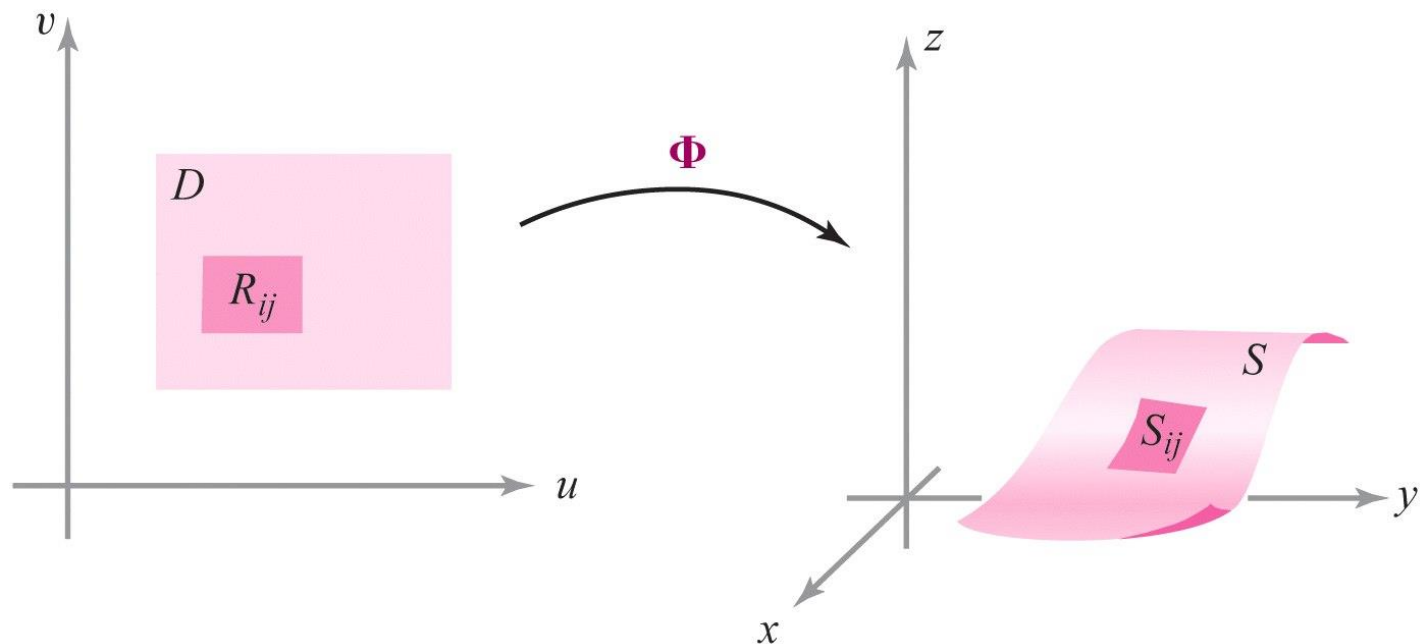
4. Surface integrals are independent of the parametrization of the surface chosen (this is discussed in the next section).
5. **Interpretation.** The total mass of a surface with a surface mass density m (mass per unit area) is given by

$$M(S) = \iint_S m(x, y, z) dS.$$

DEFINITION: The Integral of a Scalar Function Over a Surface If $f(x, y, z)$ is a real-valued continuous function defined on a parametrized surface S , we define the *integral of f over S* to be

$$\iint_S f(x, y, z) dS = \iint_S f dS = \iint_D f(\Phi(u, v)) \|\mathbf{T}_u \times \mathbf{T}_v\| du dv. \quad (1)$$

$$\|\mathbf{T}_u \times \mathbf{T}_v\| = \sqrt{\left[\frac{\partial(x, y)}{\partial(u, v)} \right]^2 + \left[\frac{\partial(y, z)}{\partial(u, v)} \right]^2 + \left[\frac{\partial(x, z)}{\partial(u, v)} \right]^2}$$



$$S_n = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} f(\Phi(u_i, v_j)) A(S_{ij})$$

$$A(S_{ij}) = \iint_{R_{ij}} \|\mathbf{T}_u \times \mathbf{T}_v\| \, du \, dv = \|\mathbf{T}_{u_i^*} \times \mathbf{T}_{v_j^*}\| \Delta u \Delta v$$

$$\lim_{n \rightarrow \infty} S_n = \iint_S f \, dS = \iint_D f(\Phi(u, v)) \|\mathbf{T}_u \times \mathbf{T}_v\| \, du \, dv$$

Surface Integrals Over Graphs

Suppose S is the graph of a C^1 function $z = g(x, y)$. Recall from Section 7.4 that we can parametrize S by

$$x = u, \quad y = v, \quad z = g(u, v),$$

and that in this case

$$\|\mathbf{T}_u \times \mathbf{T}_v\| = \sqrt{1 + \left(\frac{\partial g}{\partial u}\right)^2 + \left(\frac{\partial g}{\partial v}\right)^2},$$

so

$$\iint_S f(x, y, z) \, dS = \iint_D f(x, y, g(x, y)) \sqrt{1 + \left(\frac{\partial g}{\partial x}\right)^2 + \left(\frac{\partial g}{\partial y}\right)^2} \, dx \, dy$$

Let S be the surface defined by $z = x^2 + y$, where D is the region $0 \leq x \leq 1$, $-1 \leq y \leq 1$. Evaluate $\iint_S x \, dS$.

If we let $z = g(x, y) = x^2 + y$, formula (4) gives

$$\begin{aligned}\iint_S x \, dS &= \iint_D x \sqrt{1 + \left(\frac{\partial g}{\partial x}\right)^2 + \left(\frac{\partial g}{\partial y}\right)^2} \, dx \, dy = \int_{-1}^1 \int_0^1 x \sqrt{1 + 4x^2 + 1} \, dx \, dy \\&= \frac{1}{8} \int_{-1}^1 \left[\int_0^1 (2 + 4x^2)^{1/2} (8x \, dx) \right] dy = \frac{2}{3} \cdot \frac{1}{8} \int_{-1}^1 [(2 + 4x^2)^{3/2}] \Big|_0^1 dy \\&= \frac{1}{12} \int_{-1}^1 (6^{3/2} - 2^{3/2}) \, dy = \frac{1}{6} (6^{3/2} - 2^{3/2}) = \sqrt{6} - \frac{\sqrt{2}}{3} \\&= \sqrt{2} \left(\sqrt{3} - \frac{1}{3} \right). \quad \blacktriangle\end{aligned}$$

Evaluate $\iint_S z^2 dS$, where S is the unit sphere $x^2 + y^2 + z^2 = 1$.

For this problem, it is convenient to use spherical coordinates and to represent the sphere parametrically by the equations $x = \cos \theta \sin \phi$, $y = \sin \theta \sin \phi$, $z = \cos \phi$, over the region D in the $\theta\phi$ plane given by the inequalities $0 \leq \phi \leq \pi$, $0 \leq \theta \leq 2\pi$. From equation (1) we get

$$\iint_S z^2 dS = \iint_D (\cos \phi)^2 \|\mathbf{T}_\theta \times \mathbf{T}_\phi\| d\theta d\phi.$$

A little computation [use formula (2) of Section 7.4; see Exercise 12] shows that

$$\|\mathbf{T}_\theta \times \mathbf{T}_\phi\| = \sin \phi.$$

(Note that for $0 \leq \phi \leq \pi$, we have $\sin \phi \geq 0$.) Thus,

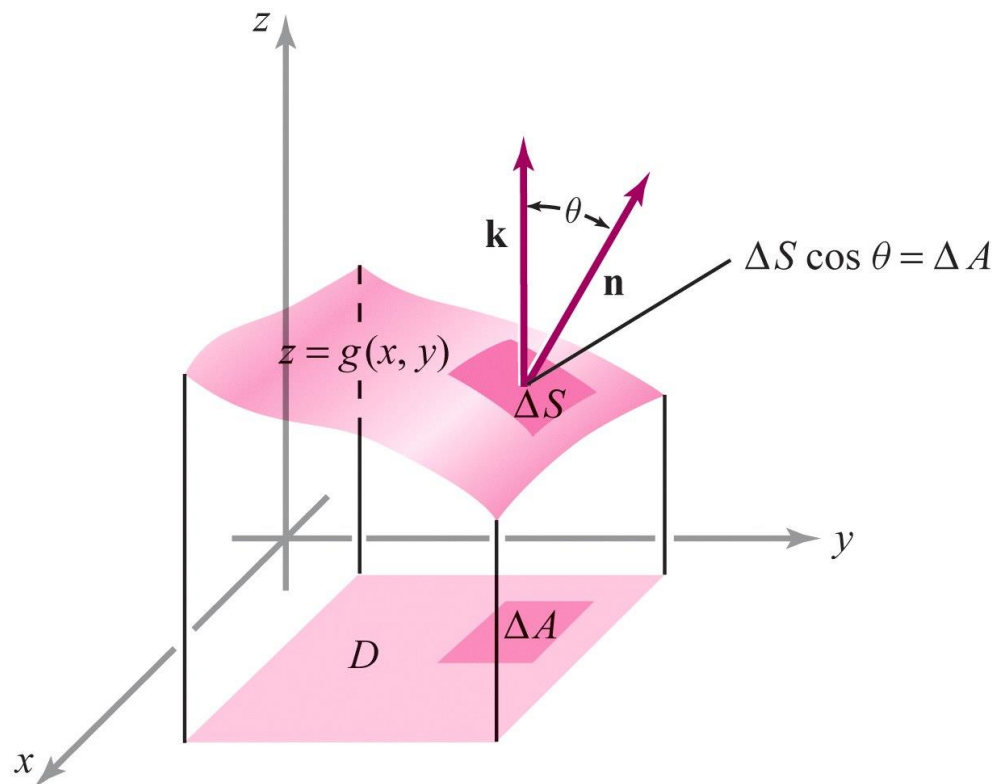
$$\begin{aligned} \iint_S z^2 dS &= \int_0^{2\pi} \int_0^\pi \cos^2 \phi \sin \phi d\phi d\theta \\ &= \frac{1}{3} \int_0^{2\pi} [-\cos^3 \phi]_0^\pi d\theta = \frac{2}{3} \int_0^{2\pi} d\theta = \frac{4\pi}{3}. \end{aligned}$$


Integrals Over Graphs

We now develop another formula for surface integrals when the surface can be represented as a graph. To do so, we let S be the graph of $z = g(x, y)$ and consider formula (4). We claim that

$$\iint_S f(x, y, z) \, dS = \iint_D \frac{f(x, y, g(x, y))}{\cos \theta} \, dx \, dy, \quad (5)$$

where θ is the angle the normal to the surface makes with the unit vector $\mathbf{k}$ at the point $(x, y, g(x, y))$



Integrals Over Graphs

We now develop another formula for surface integrals when the surface can be represented as a graph. To do so, we let S be the graph of $z = g(x, y)$ and consider formula (4). We claim that

$$\iint_S f(x, y, z) \, dS = \iint_D \frac{f(x, y, g(x, y))}{\cos \theta} \, dx \, dy, \quad (5)$$

where θ is the angle the normal to the surface makes with the unit vector $\mathbf{k}$ at the point $(x, y, g(x, y))$ (see Figure 7.5.2). Describing the surface by the equation $\phi(x, y, z) = z - g(x, y) = 0$, a normal vector $\mathbf{N}$ is $\nabla \phi$; that is,

$$\mathbf{N} = -\frac{\partial g}{\partial x} \mathbf{i} - \frac{\partial g}{\partial y} \mathbf{j} + \mathbf{k} \quad (6)$$

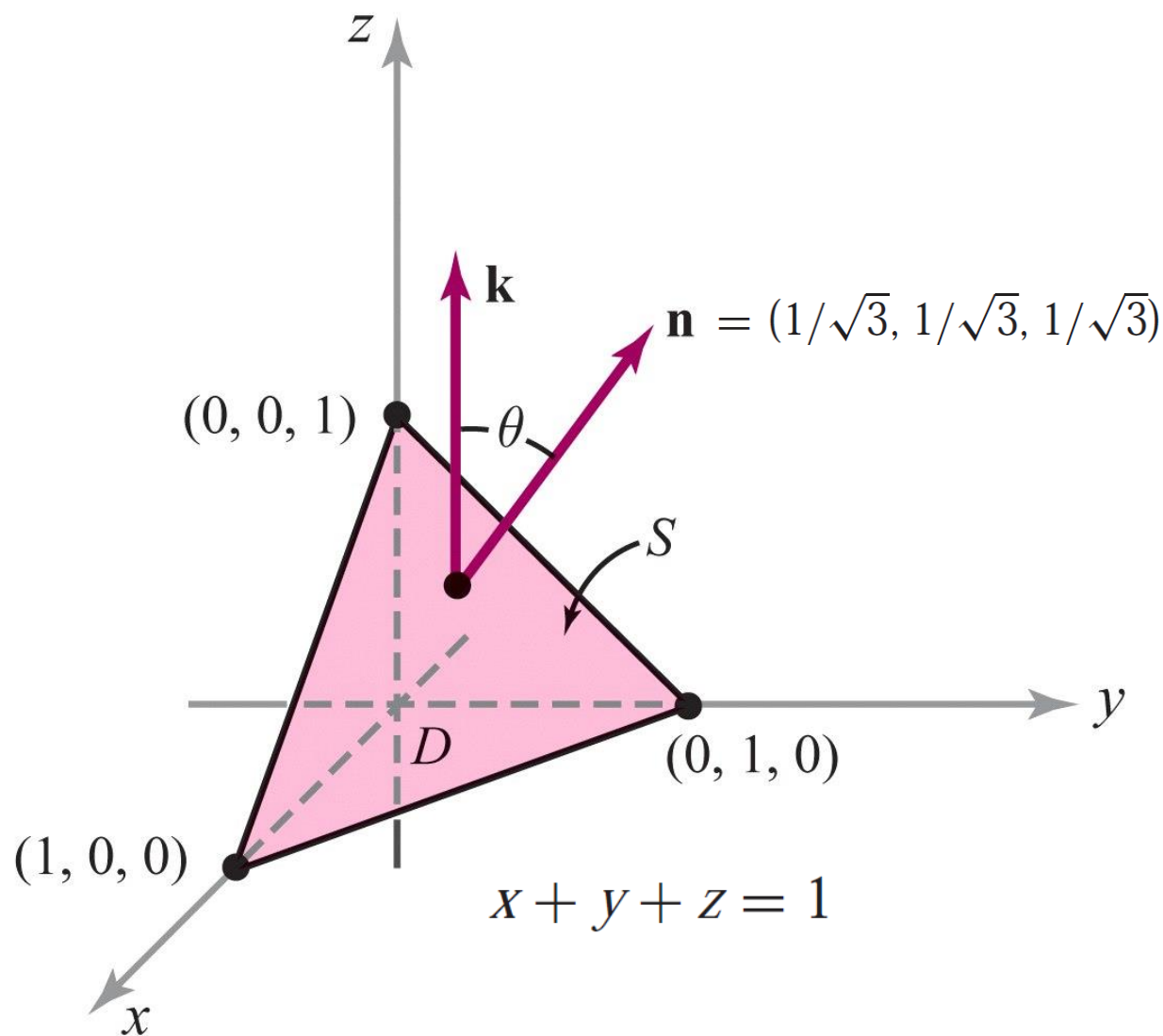
[see Example 4 of Section 7.3, or recall that the normal to a surface with equation $g(x, y, z) = \text{constant}$ is given by ∇g]. Thus,

$$\cos \theta = \frac{\mathbf{N} \cdot \mathbf{k}}{\|\mathbf{N}\|} = \frac{1}{\sqrt{(\partial g / \partial x)^2 + (\partial g / \partial y)^2 + 1}}.$$

Substitution of this formula into equation (4) gives equation (5). Note that $\cos \theta = \mathbf{n} \cdot \mathbf{k}$, where $\mathbf{n} = \mathbf{N} / \|\mathbf{N}\|$ is the unit normal. Thus, we can write

$$dS = \frac{dx \, dy}{\mathbf{n} \cdot \mathbf{k}}.$$

$$\iint_S x \, dS = \sqrt{3} \iint_D x \, dx \, dy = \sqrt{3} \int_0^1 \int_0^{1-x} x \, dy \, dx = \sqrt{3} \int_0^1 x(1-x) \, dx = \frac{\sqrt{3}}{6}$$



Jerrold E. Marsden and Anthony J. Tromba

Vector Calculus

Fifth Edition

Chapter 7:

Integrals over Paths and Surfaces

7.6 Surface Integrals of Vector Fields

7.6 Surface Integrals of Vector Functions

Key Points in this Section.

1. **Definition.** The formula for the surface integral of a vector field $\mathbf{F}$ over a parametrized surface is given by:

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F} \cdot (\mathbf{T}_u \times \mathbf{T}_v) du dv$$

2. The $d\mathbf{S}$ and dS notation helps one remember the formulas for integrals of scalar and vector functions on surfaces.
3. **Surface Area Elements—Parametrized Surface.**

$$d\mathbf{S} = \mathbf{T}_u \times \mathbf{T}_v du dv, \quad dS = \|\mathbf{T}_u \times \mathbf{T}_v\| du dv$$

or, in other notation,

$$d\mathbf{S} = \Phi_u \times \Phi_v du dv, \quad dS = \|\Phi_u \times \Phi_v\| du dv.$$

4. **Vector vs Scalar Surface Element.** Since the unit normal is $\mathbf{n} = (\mathbf{T}_u \times \mathbf{T}_v) / \|\mathbf{T}_u \times \mathbf{T}_v\|$, it follows from the preceding points that

$$d\mathbf{S} = \mathbf{n} dS.$$

5. **Vector Surface Element for a Sphere of Radius R :**

$$d\mathbf{S} = (x\mathbf{i} + y\mathbf{j} + z\mathbf{k})R \sin \phi \, d\phi \, d\theta = \mathbf{r}R \sin \phi \, d\phi \, d\theta$$

6. **Geometric Surface.** This is similar to the geometric curve idea met in line integrals. To integrate over a geometric surface, we need an orientation, or handedness. This is done by specifying a direction for the unit normal.
7. **Möbius Band.** Many students are fascinated by the fact that the Möbius band cannot be oriented. A classroom demonstration of this may be useful.

8. **Graphs.** If S is a graph $z = g(x, y)$, the default orientation is the *upward* normal. In the case of graphs, many students will want to memorize the formula

$$d\mathbf{S} = \left(-\frac{\partial g}{\partial x} \mathbf{i} - \frac{\partial g}{\partial y} \mathbf{j} + \mathbf{k} \right) dx dy,$$

which is just $\Phi_x \times \Phi_y dx dy$ where $\Phi(x, y) = (x, y, g(x, y))$.

9. **Independence of Parametrization.** As long as the orientation is respected, the surface integral over a geometric surface is well defined, independent of the parametrization. That is, for two parametrizations Φ_1 and Φ_2 , describing the same geometric surface (including the orientation), then

$$\iint_{\Phi_1} \mathbf{F} \cdot d\mathbf{S} = \iint_{\Phi_2} \mathbf{F} \cdot d\mathbf{S}.$$

Their common value is denoted

$$\iint_S \mathbf{F} \cdot d\mathbf{S}.$$

10. **Normal Component.** Since $d\mathbf{S} = \mathbf{n} dS$, we find that

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S (\mathbf{F} \cdot \mathbf{n}) dS,$$

that is, the surface integral of the vector function $\mathbf{F}$ is equal to the scalar integral of the normal component of $\mathbf{F}$.

11. **Physical Interpretation.** If $\mathbf{F}$ represents the velocity field of a fluid, then the surface integral

$$\iint_S \mathbf{F} \cdot d\mathbf{S}.$$

represents the *rate of flow* of fluid across the surface. For example, one can talk about an imaginary surface across a creek, where the flow rate might be measured in cubic meters per second. For other vector fields, the surface integral is called the *flux*. Figure 7.6.1 indicates why the flux is the integral of the normal component.

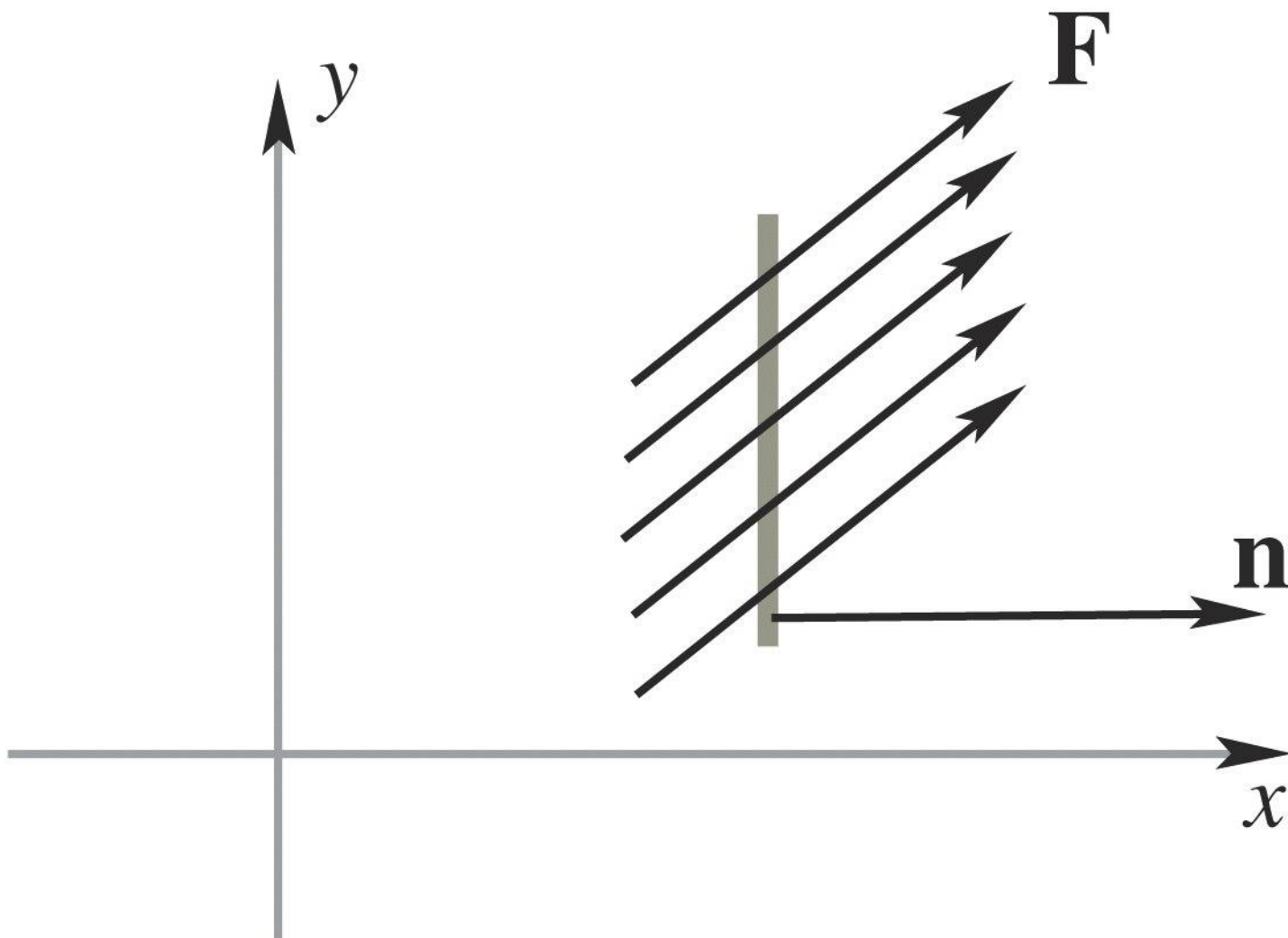


FIGURE 7.6.1. The flux across a surface (a line in two dimensions) is the integral of the normal component of the vector field.

12. **Gauss' Law.** This says (in appropriate units) that

$$\iint_S \mathbf{E} \cdot d\mathbf{S} = Q,$$

where $\mathbf{E}$ is the electric field caused by a charge distribution and Q is the total charge enclosed by the surface S .

13. **Coulomb's law.** If the charge is symmetrically placed, S is chosen to be a sphere, and one assumes (as is reasonable) that the electric field is $\mathbf{E} = E\mathbf{n}$, then one finds that

$$E = \frac{Q}{4\pi R^2}$$

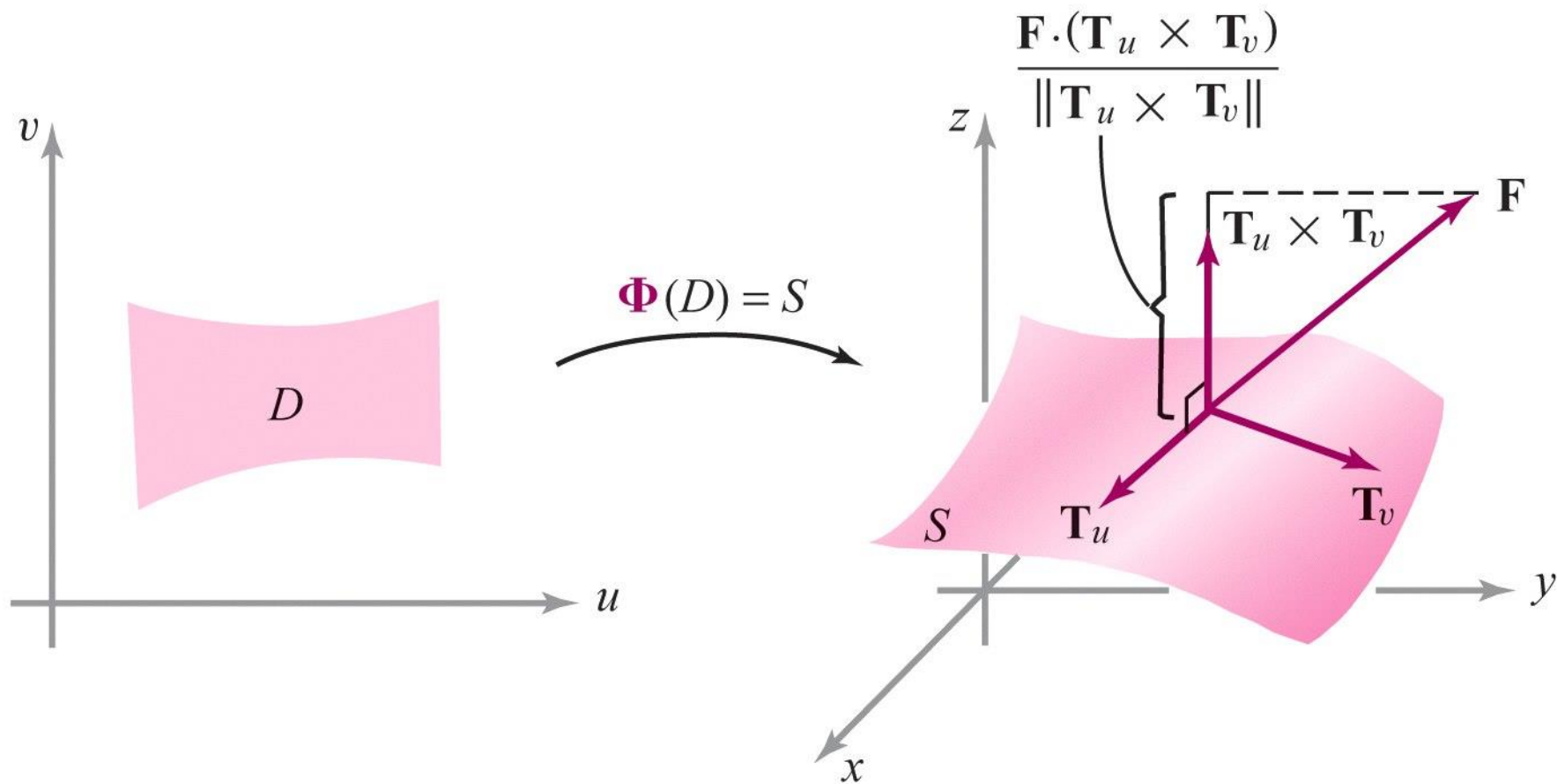
and in particular, for a point charge, one gets *Coulomb's law* stating that the above gives a formula for the field of a point charge.

DEFINITION: The Surface Integral of Vector Fields Let $\mathbf{F}$ be a vector field defined on S , the image of a parametrized surface Φ . The *surface integral* of $\mathbf{F}$ over Φ , denoted by

$$\iint_{\Phi} \mathbf{F} \cdot d\mathbf{S},$$

is defined by (see Figure 7.6.1))

$$\iint_{\Phi} \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F} \cdot (\mathbf{T}_u \times \mathbf{T}_v) du dv.$$



$$\mathbf{F} \cdot d\mathbf{S} = \mathbf{F} \cdot \mathbf{n} dS = \mathbf{F} \cdot \mathbf{n} \|\mathbf{T}_u \times \mathbf{T}_v\| du dv$$

$$= \mathbf{F} \cdot \frac{\mathbf{T}_u \times \mathbf{T}_v}{\|\mathbf{T}_u \times \mathbf{T}_v\|} \|\mathbf{T}_u \times \mathbf{T}_v\| du dv = \mathbf{F} \cdot (\mathbf{T}_u \times \mathbf{T}_v) du dv$$

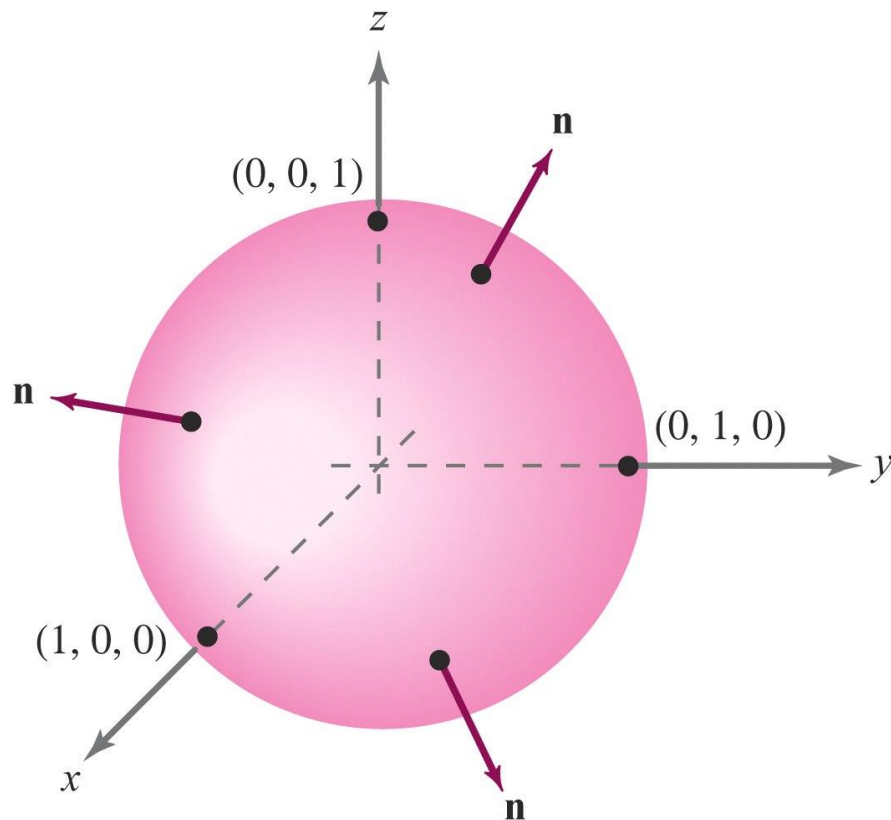
Let D be the rectangle in the $\theta\phi$ plane defined by

$$0 \leq \theta \leq 2\pi, \quad 0 \leq \phi \leq \pi,$$

and let the surface S be defined by the parametrization $\Phi: D \rightarrow \mathbb{R}^3$ given by

$$x = \cos \theta \sin \phi, \quad y = \sin \theta \sin \phi, \quad z = \cos \phi.$$

(Thus, θ and ϕ are the angles of spherical coordinates, and S is the unit sphere parametrized by Φ .) Let $\mathbf{r}$ be the position vector $\mathbf{r}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Compute $\iint_{\Phi} \mathbf{r} \cdot d\mathbf{S}$.



First we find

$$\begin{aligned}\mathbf{T}_\theta &= (-\sin \phi \sin \theta)\mathbf{i} + (\sin \phi \cos \theta)\mathbf{j} \\ \mathbf{T}_\phi &= (\cos \theta \cos \phi)\mathbf{i} + (\sin \theta \cos \phi)\mathbf{j} - (\sin \phi)\mathbf{k},\end{aligned}$$

and hence

$$\mathbf{T}_\theta \times \mathbf{T}_\phi = (-\sin^2 \phi \cos \theta)\mathbf{i} - (\sin^2 \phi \sin \theta)\mathbf{j} - (\sin \phi \cos \phi)\mathbf{k}.$$

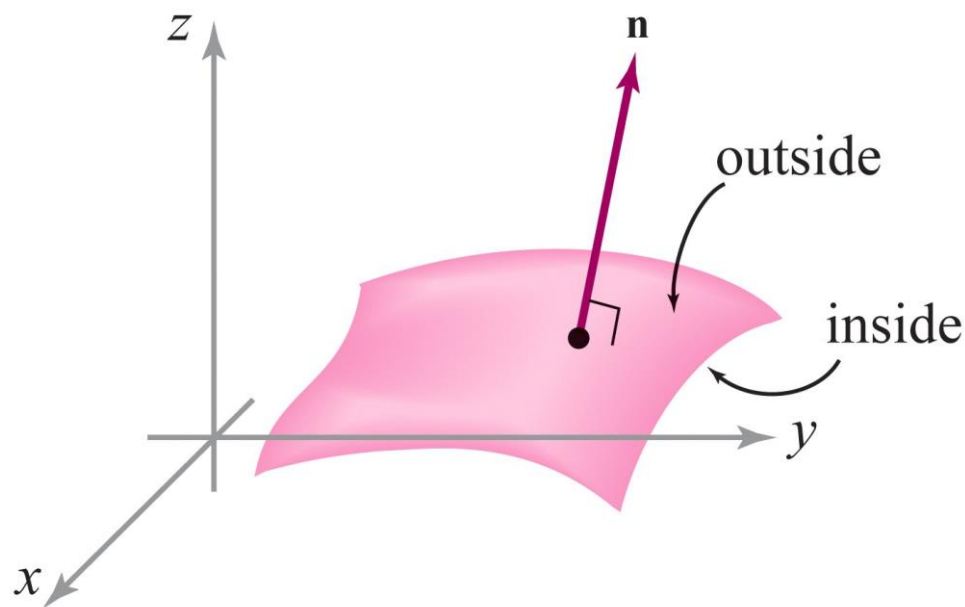
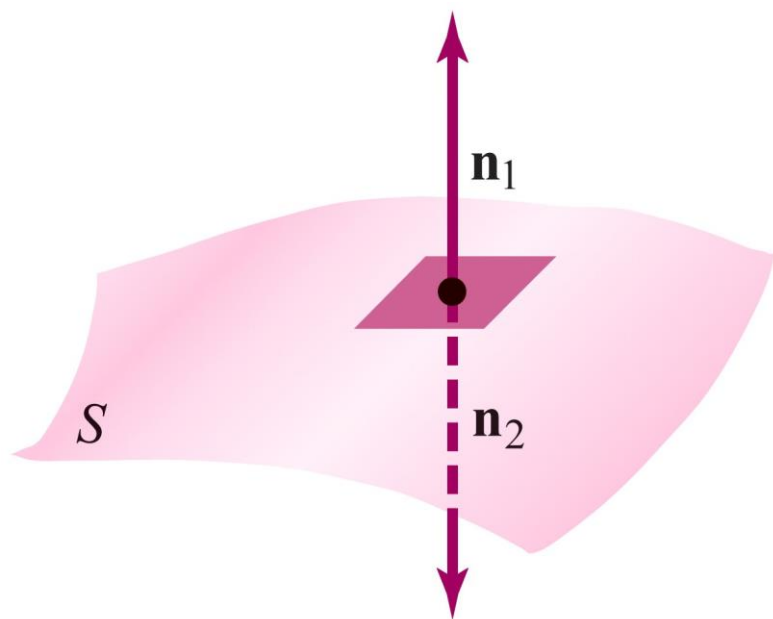
Then we evaluate

$$\begin{aligned}\mathbf{r} \cdot (\mathbf{T}_\theta \times \mathbf{T}_\phi) &= (x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) \cdot (\mathbf{T}_\theta \times \mathbf{T}_\phi) \\ &= [(\cos \theta \sin \phi)\mathbf{i} + (\sin \theta \sin \phi)\mathbf{j} + (\cos \phi)\mathbf{k}] \\ &\quad \cdot (-\sin \phi) [(\sin \phi \cos \theta)\mathbf{i} + (\sin \phi \sin \theta)\mathbf{j} + (\cos \phi)\mathbf{k}] \\ &= (-\sin \phi) (\sin^2 \phi \cos^2 \theta + \sin^2 \phi \sin^2 \theta + \cos^2 \phi) = -\sin \phi.\end{aligned}$$

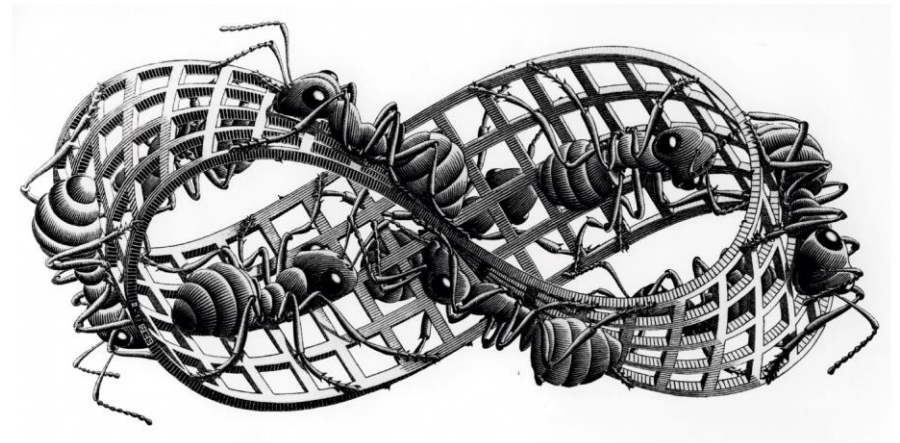
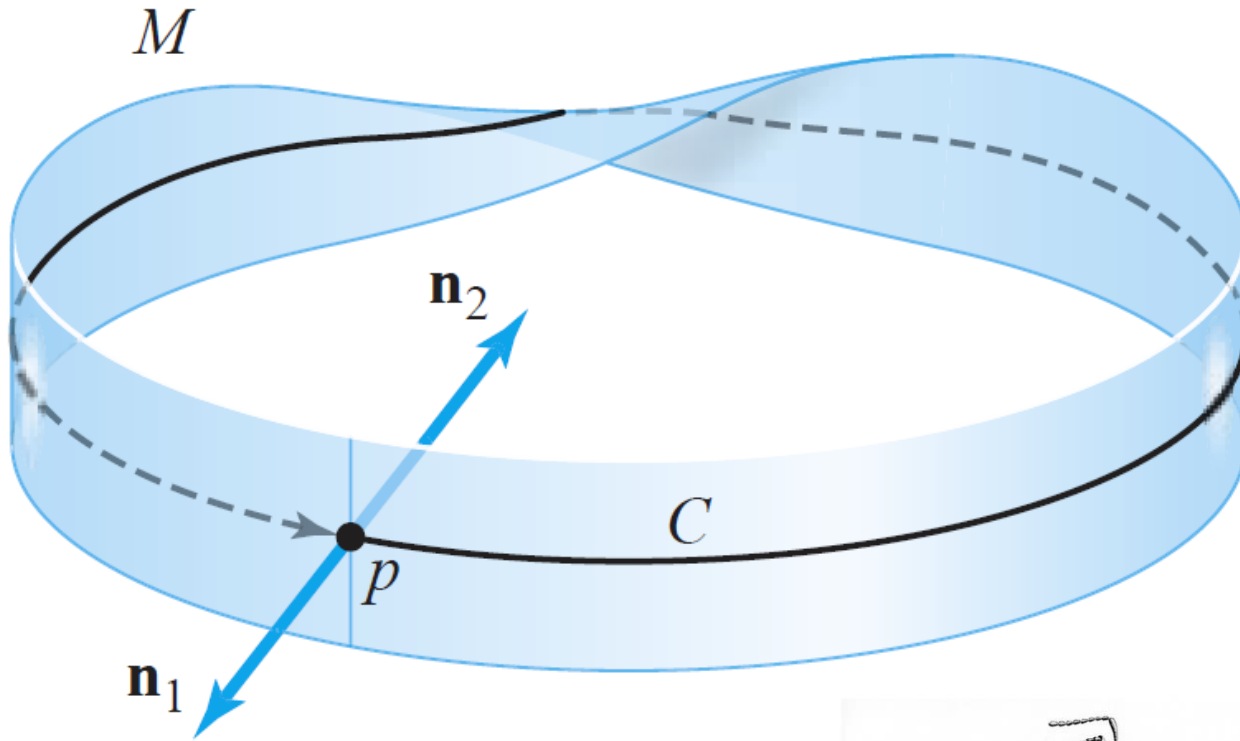
Thus,

$$\iint_{\Phi} \mathbf{r} \cdot d\mathbf{S} = \iint_D -\sin \phi \, d\phi \, d\theta = \int_0^{2\pi} (-2) \, d\theta = -4\pi. \quad \blacktriangle$$

DEFINITION: Oriented Surfaces An *oriented surface* is a two-sided surface with one side specified as the *outside* or *positive side*; we call the other side the *inside* or *negative side*.¹³ At each point $(x, y, z) \in S$ there are two unit normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$, where $\mathbf{n}_1 = -\mathbf{n}_2$ (see Figure 7.6.2). Each of these two normals can be associated with one side of the surface. Thus, to specify a side of a surface S , at each point we choose a unit normal vector $\mathbf{n}$ that points away from the positive side of S at that point.



Cinta de Möbius: superficie no orientable, només té un costat



Let $\Phi: D \rightarrow \mathbb{R}^3$ be a parametrization of an oriented surface S and suppose S is regular at $\Phi(u_0, v_0)$, $(u_0, v_0) \in D$; thus, the vector $(\mathbf{T}_{u_0} \times \mathbf{T}_{v_0}) / \|\mathbf{T}_{u_0} \times \mathbf{T}_{v_0}\|$ is defined. If $\mathbf{n}(\Phi(u_0, v_0))$ denotes the unit normal to S at $\Phi(u_0, v_0)$, it follows that

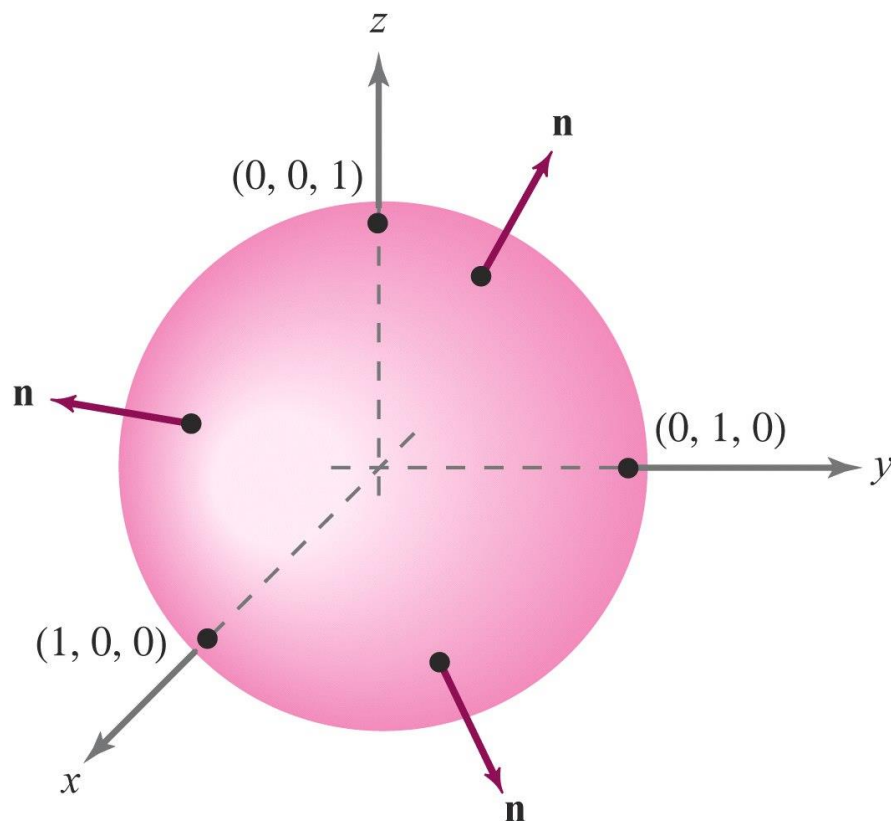
$$(\mathbf{T}_{u_0} \times \mathbf{T}_{v_0}) / \|\mathbf{T}_{u_0} \times \mathbf{T}_{v_0}\| = \pm \mathbf{n}(\Phi(u_0, v_0)).$$

The parametrization Φ is said to be *orientation-preserving* if we have the $+$ sign; that is, if $(\mathbf{T}_u \times \mathbf{T}_v) / \|\mathbf{T}_u \times \mathbf{T}_v\| = \mathbf{n}(\Phi(u, v))$ at all $(u, v) \in D$ for which S is smooth at $\Phi(u, v)$. In other words, Φ is orientation-preserving if the vector $\mathbf{T}_u \times \mathbf{T}_v$ points to the outside of the surface. If $\mathbf{T}_u \times \mathbf{T}_v$ points to the inside of the surface at all points $(u, v) \in D$ for which S is regular at $\Phi(u, v)$, then Φ is said to be *orientation-reversing*. Using the preceding notation, this condition corresponds to the choice $(\mathbf{T}_u \times \mathbf{T}_v) / \|\mathbf{T}_u \times \mathbf{T}_v\| = -\mathbf{n}(\Phi(u, v))$.

En l'exemple de l'esfera anterior, la parametrització $\theta\phi$ no preservava l'orientació ja que els vectors

$$\mathbf{T}_\theta \times \mathbf{T}_\phi = (-\sin^2\phi \cos\theta)\mathbf{i} - (\sin^2\phi \sin\theta)\mathbf{j} - (\sin\phi \cos\phi)\mathbf{k}$$

apunten cap a l'interior de l'esfera. Es podria preservar l'orientació canviant l'ordre dels paràmetres de la parametrització, és dir, amb parametrització $\phi\theta$ en lloc de $\theta\phi$.



En l'exemple de l'esfera anterior, la parametrització $\theta\phi$ no preservava l'orientació ja que els vectors

$$\mathbf{T}_\theta \times \mathbf{T}_\phi = (-\sin^2 \phi \cos \theta)\mathbf{i} - (\sin^2 \phi \sin \theta)\mathbf{j} - (\sin \phi \cos \phi)\mathbf{k}$$

apunten cap a l'interior de l'esfera. Es podria preservar l'orientació canviant l'ordre dels paràmetres de la parametrització, és dir, amb parametrització $\phi\theta$ en lloc de $\theta\phi$.

També es pot calcular la integral utilitzant directament el vector $\mathbf{n}$ correctament orientat:

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}, \quad R = \sqrt{x^2 + y^2 + z^2}$$

$$\mathbf{n} = \frac{\mathbf{r}}{R}$$

$$d\mathbf{S} = \mathbf{n} \|\mathbf{T}_\phi \times \mathbf{T}_\theta\| d\phi d\theta = \mathbf{r} R \sin \phi d\phi d\theta = \mathbf{n} R^2 \sin \phi d\phi d\theta$$

THEOREM 4: Independence of Surface Integrals on Parametrizations Let S be an oriented surface and let Φ_1 and Φ_2 be two regular orientation-preserving parametrizations, with $\mathbf{F}$ a continuous vector field defined on S . Then

$$\iint_{\Phi_1} \mathbf{F} \cdot d\mathbf{S} = \iint_{\Phi_2} \mathbf{F} \cdot d\mathbf{S}.$$

If Φ_1 is orientation-preserving and Φ_2 orientation-reversing, then

$$\iint_{\Phi_1} \mathbf{F} \cdot d\mathbf{S} = - \iint_{\Phi_2} \mathbf{F} \cdot d\mathbf{S}.$$

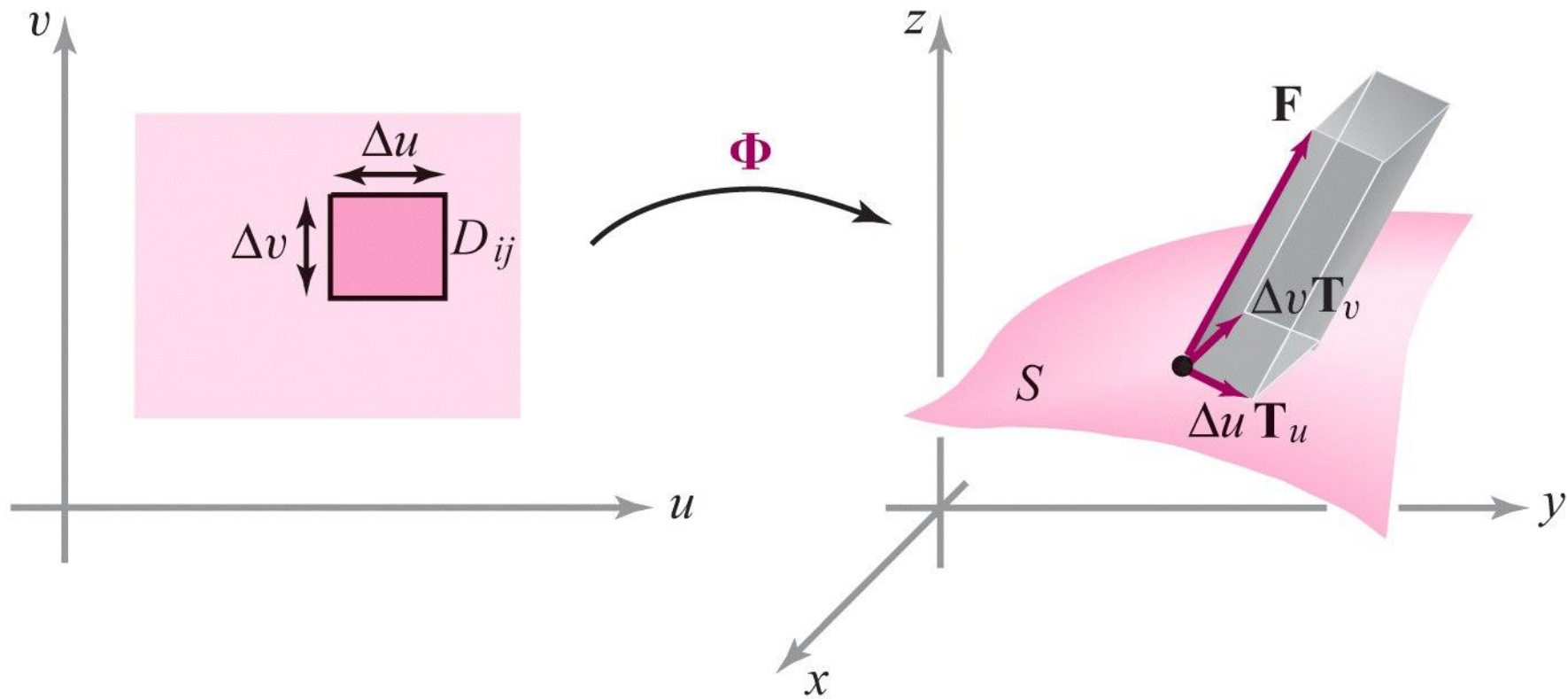
If f is a real-valued continuous function defined on S , and if Φ_1 and Φ_2 are parametrizations of S , then

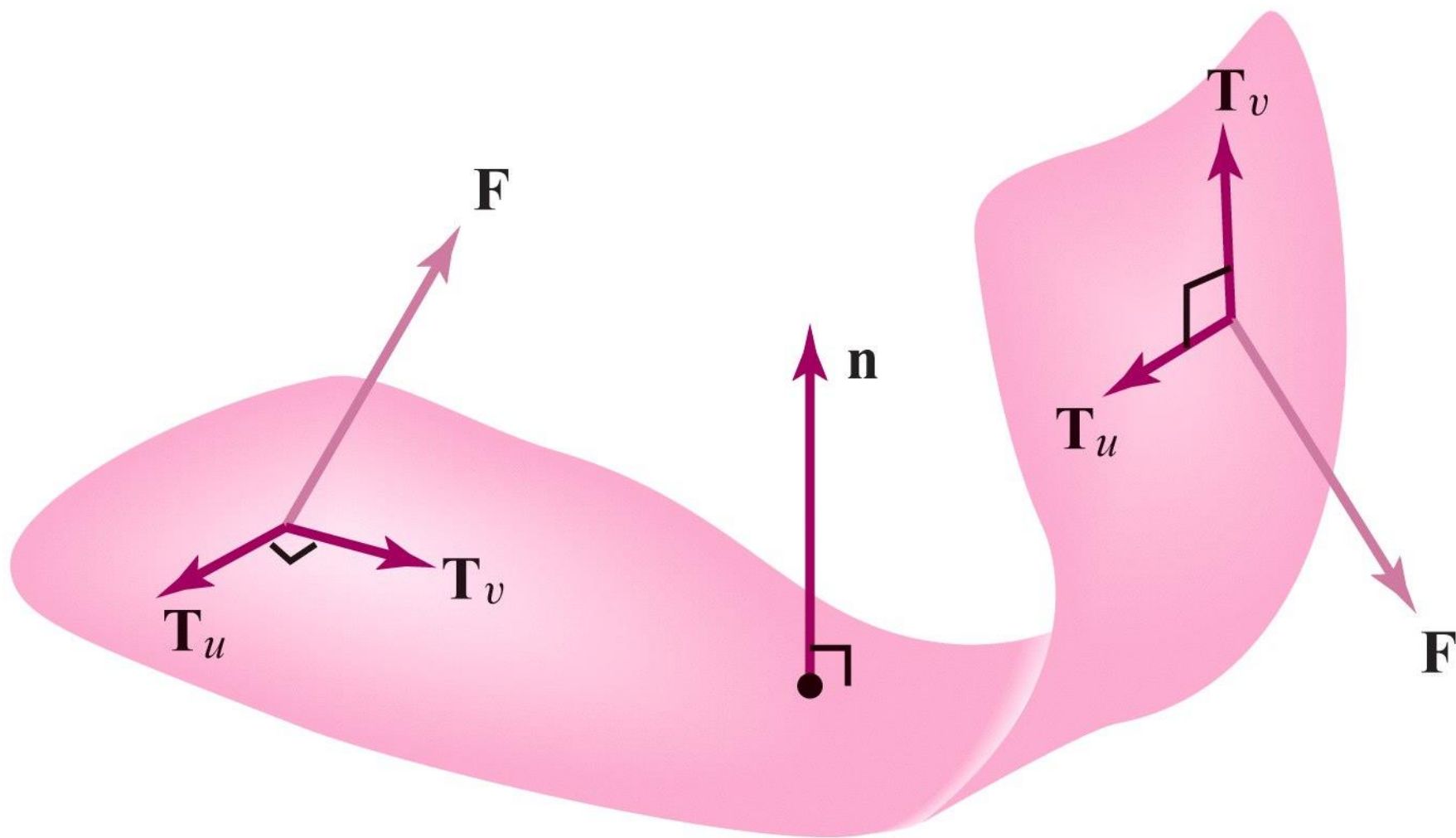
$$\iint_{\Phi_1} f dS = \iint_{\Phi_2} f dS.$$

THEOREM 5 $\iint_S \mathbf{F} \cdot d\mathbf{S}$, the surface integral of $\mathbf{F}$ over S , is equal to the integral of the normal component of $\mathbf{F}$ over the surface. In short,

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot \mathbf{n} \, dS.$$

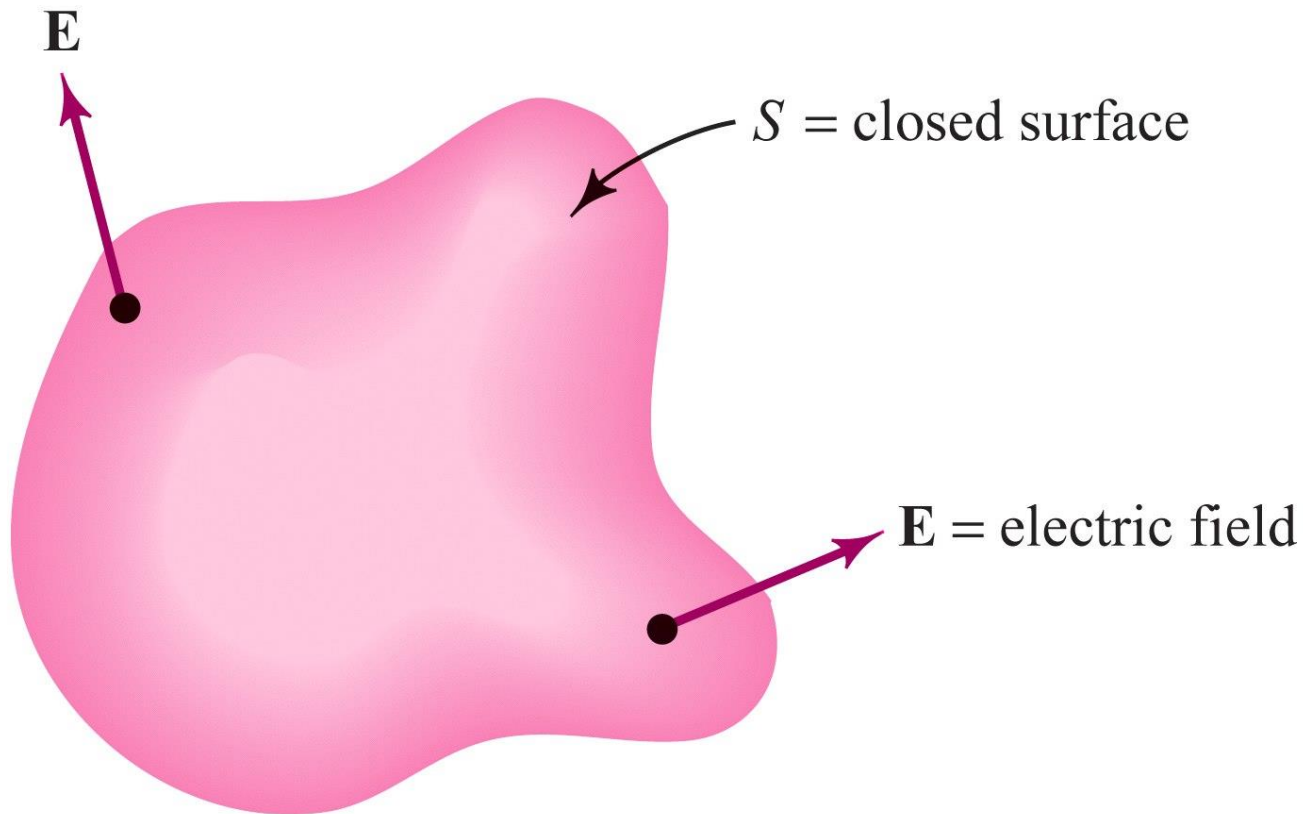
$$\begin{aligned} \iint_S \mathbf{F} \cdot d\mathbf{S} &= \iint_{\Phi} \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F} \cdot (\mathbf{T}_u \times \mathbf{T}_v) \, du \, dv \\ &= \iint_D \mathbf{F} \cdot \left(\frac{\mathbf{T}_u \times \mathbf{T}_v}{\|\mathbf{T}_u \times \mathbf{T}_v\|} \right) \|\mathbf{T}_u \times \mathbf{T}_v\| \, du \, dv \\ &= \iint_D (\mathbf{F} \cdot \mathbf{n}) \|\mathbf{T}_u \times \mathbf{T}_v\| \, du \, dv = \iint_S (\mathbf{F} \cdot \mathbf{n}) \, dS \end{aligned}$$





Llei de Gauss del flux del camp elèctric

$$\iint_S \mathbf{E} \cdot d\mathbf{S} = Q$$



The Surface Integral of a Vector Field Over a Graph S $z = g(x, y)$

$$\begin{aligned}\iint_S \mathbf{F} \cdot d\mathbf{S} &= \iint_D \mathbf{F} \cdot (\mathbf{T}_x \times \mathbf{T}_y) dx dy \\ &= \iint_D \left[F_1 \left(-\frac{\partial g}{\partial x} \right) + F_2 \left(-\frac{\partial g}{\partial y} \right) + F_3 \right] dx dy.\end{aligned}\tag{4}$$

$$\boldsymbol{\phi}(x, y, z) = (x, y, g(x, y))$$

$$\mathbf{T}_x = \mathbf{i} + \frac{\partial g}{\partial x} \mathbf{k}$$

$$\mathbf{T}_y = \mathbf{j} + \frac{\partial g}{\partial y} \mathbf{k}$$

$$\mathbf{T}_x \times \mathbf{T}_y = -\frac{\partial g}{\partial x} \mathbf{i} - \frac{\partial g}{\partial y} \mathbf{j} + \mathbf{k}$$

The equations

$$z = 12, \quad x^2 + y^2 \leq 25$$

describe a disc of radius 5 lying in the plane $z = 12$. Suppose $\mathbf{r}$ is the vector field

$$\mathbf{r}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}.$$

Compute $\iint_S \mathbf{r} \cdot d\mathbf{S}$.

We shall do this in three ways. First, we have $\partial z/\partial x = \partial z/\partial y = 0$, because $z = 12$ is constant on the disc, so

$$\mathbf{r}(x, y, z) \cdot (\mathbf{T}_x \times \mathbf{T}_y) = \mathbf{r}(x, y, z) \cdot (\mathbf{i} \times \mathbf{j}) = \mathbf{r}(x, y, z) \cdot \mathbf{k} = z.$$

Using the original definition at the beginning of this section, the integral becomes

$$\iint_S \mathbf{r} \cdot d\mathbf{S} = \iint_D z \, dx \, dy = \iint_D 12 \, dx \, dy = 12(\text{area of } D) = 300\pi.$$

A second solution: Because the disc is parallel to the xy plane, the outward unit normal is $\mathbf{k}$. Hence, $\mathbf{n}(x, y, z) = \mathbf{k}$ and $\mathbf{r} \cdot \mathbf{n} = z$. However, $\|\mathbf{T}_x \times \mathbf{T}_y\| = \|\mathbf{k}\| = 1$, and so we know from the discussion preceding Theorem 5 that

$$\iint_S \mathbf{r} \cdot d\mathbf{S} = \iint_S \mathbf{r} \cdot \mathbf{n} \, dS = \iint_S z \, dS = \iint_D 12 \, dx \, dy = 300\pi.$$

Third, we may solve this problem by using formula (4) directly, with $g(x, y) = 12$ and D the disc $x^2 + y^2 \leq 25$:

$$\iint_S \mathbf{r} \cdot d\mathbf{S} = \iint_D (x \cdot 0 + y \cdot 0 + 12) \, dx \, dy = 12(\text{area of } D) = 300\pi. \quad \blacktriangle$$